Beyond the Status Quo, Revisited

A Computational Re-Examination of Lifecycle Asset Allocation, with Ten Extensions

A replication and extension study

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Abstract

We reproduce and extend the central result of Anarkulova, Cederburg and O'Doherty (2023, 2024): that a lifecycle investor holding only equities, split evenly between domestic and international markets, delivers higher certainty-equivalent retirement consumption than the age-declining glide path embedded in target-date funds. Working from a 16-country developed-market panel of real asset returns spanning 1890–2020 (2,010 country-years), we build a calendar-joint stationary block bootstrap that samples whole (country, window) blocks and therefore preserves cross-asset covariance, long-horizon persistence and the fat tails of the historical record. Simulating 100,000 lifetimes per strategy under CRRA and Epstein–Zin preferences, we recover the qualitative result: the all-equity portfolio leads the target-date fund by 11.2% and a 60/40 portfolio by 20.3% in certainty-equivalent consumption at $\gamma = 5$, while cutting the probability of exhausting the portfolio from 22% to 12%. The ranking survives 132 parameter settings across ten dimensions with 0 reversals. Sustainable withdrawal rates are far below the four-percent convention on every strategy (3.2% for all-equity, 2.1% for the target-date fund at a five-percent ruin tolerance). We then push past replication with ten extensions. Solving the glide path directly by coordinate ascent under common random numbers reproduces the all-equity corner rather than an interior optimum, and freeing all four portfolio weights at every age — 204 parameters on the simplex — adds 0.30% over the best fixed benchmark while still producing no glide path. Relaxing the long-only constraint, borrowing to invest is worth +5.10% at a zero borrowing spread over the real bill rate but decays quickly in that spread, breaking even by 3.00% — and what the optimiser levers is a diversified portfolio rather than a concentrated one. Currency hedging the international leg loses certainty-equivalent consumption at every ratio tested, even when the hedge is free. Making the retirement date a wealth-triggered decision rather than a birthday is worth about 3% of certainty-equivalent consumption against a date matched on the same mean retirement age. Conditioning the savings rate on the funded ratio is worth a further 4.4%, and a deep decomposition of that signal — across functional form, target definition, asymmetry, feasibility bands and eight competing state variables — finds that the strongest available signal is not the portfolio at all but the investor's own pay cheque (6.4%). Finally, the decade around the retirement date explains 6% of the variation in retirement outcomes, a lottery no allocation rule can diversify away. The panel's principal limitation is its breadth: 16 developed markets, so the international leg spans 15 foreign markets, all advanced economies with long recorded histories.

Keywords: lifecycle asset allocation; target-date funds; block bootstrap; certainty-equivalent consumption; safe withdrawal rate; sequence-of-returns risk; glide path; retirement timing; savings rate; international diversification.

JEL classification: G11, G51, D14, D15, J26, C15.

Contents

1. Introduction

The default investment vehicle of the modern retirement system is the target-date fund. It embodies a single, rarely-examined proposition: that an investor should hold less equity as they age, sliding from a growth portfolio in their twenties toward a bond-heavy portfolio at retirement. That proposition is not a market observation. It is a theoretical inheritance, descended from Samuelson (1969) and Merton (1969) via the human-capital arguments of Bodie, Merton and Samuelson (1992), and it has been institutionalised across trillions of dollars of default-enrolled savings.

Anarkulova, Cederburg and O'Doherty (henceforth ACO) challenged it directly. Their argument is not that equities have higher expected returns — everyone agrees on that — but that the standard evidence against holding them is an artefact of how returns are simulated. Draw returns independently and identically distributed from a normal-ish distribution fitted to post-war US data, and equities look dangerous over short horizons and safe over long ones in exactly the way textbook time diversification predicts. Draw them instead in blocks from the full international historical record — including the markets that closed, the decades that were lost, and the countries whose real returns were negative for a generation — and the shape of the problem changes. ACO's conclusion is that a fixed all-equity portfolio, split between domestic and international markets, beats the glide path on almost every metric an investor would care about.

This paper does three things. First, it reproduces that result from scratch on an independently constructed panel, with the full diagnostic apparatus made visible rather than asserted. Second, it stress-tests the result across a large parameter space, so that the reader can see where the conclusion is robust and where it is thin. Third — and this is where most of the length lies — it takes eight extensions that the original design leaves open and works each of them out, several of which produce results that run against the intuition that motivated them.

1.1 What we find

The replication succeeds. On a 16-country panel covering 1890–2020 (2,010 country-years), the 50/50 domestic/international equity portfolio delivers 11.2% higher certainty-equivalent consumption than a target-date glide path, 20.3% higher than a 60/40 portfolio and 18.8% higher than domestic equity alone, at the baseline risk aversion $\gamma = 5$. It does so while *reducing* the probability of running out of money from 22.2% to 11.7%. Across the 21 distributional criteria we test, the all-equity portfolio wins 20 and loses none.

The mechanism is international diversification, not equity risk per se. Domestic-only equity is beaten by the target-date fund on several tail measures. What makes the all-equity portfolio work is that a bad domestic century and a bad international century are not the same century. This is a stronger claim than "stocks beat bonds" and it is the one the international panel is needed to make.

Four percent is not a safe withdrawal rate on this panel. At a five-percent ruin tolerance the sustainable rate is under three percent for every strategy tested, and under two percent for the conservative ones. The four-percent rule is a US-post-war artefact and does not survive a sample that contains the twentieth century as the rest of the world experienced it.

Solving the glide path directly returns the corner. Rather than testing a handful of candidate schedules, we solve for the equity share at every age by coordinate ascent under common random numbers. The solution is not an interior glide path: it sits at or near the all-equity corner for the entire lifecycle, and the deviation profile shows that most of the apparent age-structure in the solved schedule is worth less than a basis point.

Freeing every weight changes almost nothing. Solving the full four-asset simplex at every age — 204 free parameters, with the bond/bill split no longer imposed — beats the best fixed benchmark by 0.30%. The optimiser holds essentially no fixed income at any age, so the restriction the glide-path solve made for convenience turns out not to have been binding.

Leverage stops being worth the trouble long before it stops paying. Borrowing to invest is worth +5.10% when credit is free and breaks even at a spread of 3.00%, but the advantage is already under a tenth of a percent by 3.00%. What the optimiser levers is a diversified portfolio rather than a concentrated one, and the cost of the trade is paid in the left tail.

Timing beats allocation. The single decade around a person's retirement date explains more of the variation in their retirement outcome than the choice between any two of the allocation strategies we test. Making the retirement date itself a decision — retire when wealth reaches a multiple of income, rather than on a birthday — is worth roughly three percent of certainty-equivalent consumption against a fixed date matched on the same mean retirement age.

The accumulation side has more room than the allocation side. Conditioning how much you save on whether you are ahead of or behind an age-appropriate wealth target is worth more than any of the allocation refinements we test. Decomposing that signal produces the paper's most counter-intuitive result: the best state variable available to a saver is not their portfolio balance but their own labour income relative to its expected path.

1.2 What is new here

Relative to the paper being replicated, this study contributes a methodological discipline and ten substantive extensions. The discipline is a set of comparison rules applied uniformly: every policy is scored against a *matched* baseline that differs from it in exactly one dimension; every optimiser runs under common random numbers so that differences are policy effects rather than sampling noise; every solved schedule is subjected to a deviation profile before any structure in it is described; and every apparent optimum is checked against the boundary of its own search grid. Several of the results below changed sign or magnitude when these checks were applied, and the paper reports the corrected versions with the diagnostics that forced them.

- **Sensitivity.** A tornado analysis over ten parameter dimensions, reporting the full range of the advantage rather than a point estimate.
- **Retirement spending rules.** Eight families of withdrawal policy — constant real, percentage-of-portfolio, guardrails, endowment smoothing, required-minimum-distribution, amortisation, actuarial and a floor-and-ceiling rule — each optimised over its own rate and compared at its own best setting.
- **The optimal glide path.** Direct numerical solution of the age-by-asset schedule, free-form and parametric, with local-optimum checks from multiple restarts.
- **The whole allocation.** The full four-asset weight simplex solved at every year of the lifecycle — domestic equity, international equity, bonds and bills, with nothing held fixed — by two-stage coordinate ascent on the simplex.
- **Leverage.** The long-only constraint relaxed: the optimal borrowing ratio and the portfolio that goes with it, swept across the price of credit, with a break-even spread and an age-varying leverage schedule.
- **Currency hedging.** A covered-interest-parity hedged international leg, swept by annual hedging cost, yielding a break-even cost and an optimal hedge ratio as functions of that cost.
- **Endogenous retirement timing.** Retirement as a wealth-triggered decision, plus a formal decomposition of the retirement-date lottery.
- **The savings-rate profile.** Solving for the savings rate at every age with the career average pinned, which separates the shape question from the level question the model cannot answer.
- **Conditioning the savings rate.** Making the contribution respond to portfolio state, scored against a constant rate matched on the realised career average.

- **The accumulation signal, decomposed.** Functional form, target definition, response asymmetry, feasibility bands, age windows, eight competing state variables and their pairwise combination.

1.3 Roadmap

Section 2 places the exercise in the literature. Section 3 documents the panel and its construction. Section 4 sets out the bootstrap, the lifecycle model, the preference specification and the comparison discipline. Section 5 presents the baseline replication and Section 6 its sensitivity. Sections 7 through 14 present the ten extensions in order of increasing distance from the original design. Section 15 discusses what the results collectively imply, Section 16 states the limitations candidly, and Section 17 concludes. Four appendices give the full parameter set, the country panel, supplementary tables and the reproduction instructions.

2. Background and Related Work

2.1 The theoretical case for the glide path

The canonical result is negative: in the Samuelson–Merton framework with constant relative risk aversion, i.i.d. returns and no labour income, the optimal share of wealth in risky assets is independent of the investment horizon. Time does not diversify risk; it multiplies it. Any age-declining allocation therefore has to be justified by something outside that framework.

The standard justification is human capital. Bodie, Merton and Samuelson (1992) observe that a young worker holds a large stock of implicit wealth in the form of future labour income, and that if this asset is bond-like then total wealth is already heavily weighted toward safety; the financial portfolio should compensate by holding more equity. As the worker ages, human capital depletes and the financial portfolio should become more conservative to keep total exposure roughly constant. This produces a glide path, and it is the argument the target-date industry rests on.

The argument is coherent but its quantitative force depends on parameters that are difficult to pin down: how bond-like labour income actually is, how it correlates with equity markets, how much of it remains at each age, and what happens to it in exactly the states of the world where equities fail. It also depends on what the return distribution looks like — which is where the argument this paper reproduces enters.

2.2 The empirical challenge

Almost all of the simulation evidence used to calibrate glide paths draws returns from a parametric distribution fitted to a single market — usually the United States — over a period that begins after the Second World War. That sample has three properties that flatter conservative allocations in a specific direction. It excludes the market closures, expropriations and hyperinflations that other developed markets experienced. It excludes the pre-war period in which the US itself behaved less benignly. And by drawing independently it destroys the multi-decade persistence that makes long-horizon outcomes fat-tailed.

ACO's response is to replace the return-generating process. Their sample is the developed-market cross-section over more than a century, and their sampling scheme draws contiguous blocks so that runs of good and bad decades survive into the simulated lifetimes. The claim is that under this process the case for the glide path evaporates. This paper reconstructs that process independently and asks whether the claim holds.

2.3 The data source

The underlying return data come from the Jordà–Schularick–Taylor Macrohistory Database and the associated "Rate of Return on Everything" project (Jordà, Knoll, Kuvshinov, Schularick and Taylor, 2019). That database assembles annual total returns on equities, long-term government bonds, short-term bills and consumer price inflation for sixteen advanced economies from the late nineteenth century onward. It is the standard source for long-horizon international asset-return work and it is what makes an exercise of this kind possible at all.

It is also the binding constraint on this study: it carries complete equity, bond and bill total returns for 16 countries, and that set is our panel. Section 3.2 describes it and Section 16.1 sets out what its breadth costs.

2.4 Adjacent literatures this paper touches

- **Bootstrap inference for dependent data.** The sampling scheme here is a stationary block bootstrap in the sense of Politis and Romano (1994), generalised so that a block is a (country, window) pair drawn jointly across all five series.
- **Safe withdrawal rates.** The four-percent convention descends from Bengen (1994) and the Trinity study (Cooley, Hubbard and Walz, 1998), both of which are US-only and historical-sequence based. Section 5.4 re-derives the sustainable rate on this panel.
- **Dynamic withdrawal policy.** Guyton and Klinger (2006) guardrails and the actuarial/amortisation family are represented in the spending-rule comparison of Section 7.
- **Bequest motives.** The utility specification uses the shifted power form of De Nardi (2004), which is what allows a zero terminal balance to be evaluated at all under high risk aversion.
- **Sequence-of-returns risk.** Section 12 formalises the folk observation that the decade around retirement dominates, and prices the option value of choosing when to stop working.
- **Lifecycle leverage.** Ayres and Nalebuff (2010) argue that a young investor should borrow to spread market exposure evenly across a lifetime. Section 10 relaxes the long-only constraint and prices that argument against the cost of credit.

3. Data

3.1 What the panel is

The panel is an annual, country-by-year matrix of five real series — domestic equity, international equity, long-term government bonds, short-term bills and consumer price inflation — for 16 developed markets over 1890–2020. It contains 2,010 usable country-years of domestic equity returns, with a median country history of 125 years and a minimum of 115 years.

All series are *real*: nominal total returns are deflated by each country's own consumer price index in the same year, so that a hyperinflation shows up as a catastrophic real return rather than as a spectacular nominal one. This is not a cosmetic choice. Several of the worst episodes in the sample — Germany in the early 1920s, central Europe in the late 1940s — are invisible in nominal space and dominate the left tail in real space.

The equity series across the panel average 6.8% in arithmetic real terms and 4.3% geometrically, with a mean cross-country standard deviation of 22.2%. The gap between the arithmetic and geometric means — over three percentage points — is itself the story: at this level of volatility, the compounded experience of a lifetime investor is far below the average annual return, and any simulation that draws i.i.d. from the arithmetic mean will overstate what a real investor would have received.

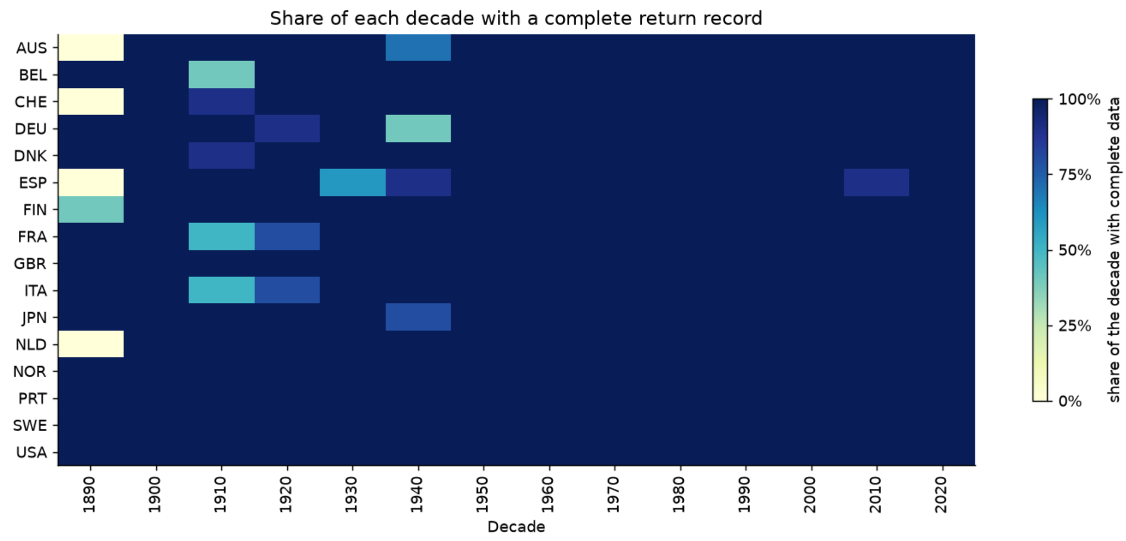

Figure 1.

Share of each decade for which a country has a complete return record. Darker is more complete; the scale is a share so that the final, partial decade compares with the rest. The gaps are not random: they cluster around the two world wars and around the market closures that follow them, which is precisely the period a survivorship-prone sample would drop.

3.2 The country set

The Jordà–Schularick–Taylor database carries complete, independently sourced equity, bond and bill total returns for 16 countries: Australia, Belgium, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, United States. That set is the panel, and every country-year of every return series in it is a recorded observation.

The wider developed-market universe — IMF advanced economies with an investable domestic equity market, plus Poland — numbers 38. The histories of the other 22 exist in commercial databases such as Global Financial Data and Dimson–Marsh–Staunton, which are proprietary and not redistributable. Working from openly licensed sources, 16 is the extent of the recorded evidence, and Section 3.6.1 reports what the excluded countries do carry.

That breadth is the paper's principal limitation and Section 16.1 develops it. Its most direct consequence is on the international leg, which is a leave-one-out average and therefore spans 15 foreign markets, all advanced economies with long histories. Every statement here about international diversification is a statement about that set.

3.3 Constructing the international leg

An investor in country *i* holding "international equity" does not hold a global index that includes their own market. For each country-year we therefore build the international return as a leave-one-out average across all other countries with data that year, computed in gross terms and in a common currency before being converted back to the home investor's real terms. Leaving the home market out is what makes the domestic and international legs genuinely distinct assets rather than two overlapping slices of the same one.

Three observations in the raw international series are extreme enough to dominate any lifetime that draws them, all of them artefacts of a market reopening after wartime closure with a collapsed price index. We winsorise the international leg at the 0.1th percentile of its own distribution; the affected observations are listed in full in Table 2 so that the reader can judge the intervention rather than take it on trust.

Table 1. Real domestic equity returns by country (first twenty by ISO code)

Country	Years	From	To	Mean	Geo. mean	S.d.	Skew	AR(1)
Australia	118	1900	2020	7.8%	6.4%	16.8%	-0.31	-0.07
Belgium	125	1890	2020	5.9%	3.8%	20.7%	-0.00	0.08
Switzerland	120	1900	2020	6.8%	5.1%	18.9%	0.30	0.10
Germany	124	1890	2020	8.7%	4.6%	28.9%	1.08	-0.01
Denmark	130	1890	2020	7.5%	6.1%	17.2%	0.43	0.04
Spain	115	1900	2020	6.2%	4.2%	21.1%	0.77	0.32
Finland	125	1896	2020	8.9%	5.0%	30.0%	1.21	0.18
France	124	1890	2020	3.6%	1.1%	23.3%	0.86	0.15
United Kingdom	131	1890	2020	6.6%	4.9%	18.7%	0.90	-0.05
Italy	124	1890	2020	6.1%	2.6%	26.8%	0.66	-0.04
Japan	129	1890	2020	7.9%	4.2%	25.8%	0.26	0.12
Netherlands	121	1900	2020	7.1%	5.1%	21.2%	0.74	0.03
Norway	131	1890	2020	5.7%	3.7%	20.1%	0.46	0.02
Portugal	131	1890	2020	3.2%	-0.1%	26.0%	0.66	0.25
Sweden	131	1890	2020	8.1%	6.1%	20.3%	0.05	0.13
United States	131	1890	2020	8.4%	6.7%	18.7%	-0.24	0.00

Notes. Every country's equity is observed; there is no other kind in this panel. The full full table is Appendix B. Arithmetic and geometric means are annual real returns; AR(1) is the first-order autocorrelation of the annual series.

3.4 Cross-asset structure

The correlation structure the bootstrap must preserve is reported below. Domestic and international equity correlate at 0.43, which is high enough that international diversification is not free and low enough that it is worth having. Bonds and bills correlate at 0.72. Inflation correlates negatively with every asset, most strongly with bills (-0.66), which is the mechanism by which cash loses in real terms over long horizons.

Table 2. Pooled cross-asset correlation matrix of annual real returns

	Dom. equity	Intl. equity	Bonds	Bills	Inflation
Domestic equity	1.000	0.429	0.315	0.234	-0.121
International equity	0.429	1.000	0.110	0.117	-0.042
Bonds	0.315	0.110	1.000	0.716	-0.469
Bills	0.234	0.117	0.716	1.000	-0.662
Inflation	-0.121	-0.042	-0.469	-0.662	1.000

Notes. Pooled across all country-years in the panel. These are the moments the block bootstrap is required to reproduce; Table 4 reports how closely it does so.

Table 3. Every winsorised observation in the international leg

Year	Country	Raw return	After winsorisation
1942	Italy	1120%	301%
1945	Japan	-86%	-70%
1946	France	308%	301%

Notes. All three are markets reopening after wartime closure against a collapsed price index. Left unwinsorised, a single one of these draws would dominate the terminal wealth of any lifetime that contained it.

3.5 Market disruptions and the survivorship question

There are 16 contiguous runs of missing data across the countries of the panel, concentrated in the two world wars. The treatment of these gaps is a substantive modelling decision rather than a data-cleaning one. Interpolating across them would smooth away exactly the episodes that motivate the whole exercise; dropping the countries would reintroduce the survivorship bias the panel exists to avoid.

We do neither. Gaps are preserved as gaps, and the bootstrap is constrained to draw only blocks that lie entirely within a contiguous run of available data for the country in question. A country with a six-year wartime hole contributes blocks on either side of it but never across it. The consequence is that the sampler is *less* able to draw from countries with disrupted histories at long block lengths, which we quantify rather than assume: the run-length admissibility statistics are reported in Section 4.1.

3.6 Auditing the data

The database is used here through a redistributed copy rather than obtained from its compilers, which deserves more scepticism than a citation. This section asks whether the file is genuine, reports one test it does not pass, and describes three bodies of recorded data that sit in the sources and stay out of the model.

3.6.1 Why the panel stops at sixteen

The boundary is set by equity. Four countries outside the panel do carry recorded interest-rate histories: long-term yields and short rates in the macro file for Canada and Ireland, Clio-Infra long yields for Austria and New Zealand. Those are enough to build real bond and bill returns — a bond total return follows from a yield and a duration, $r_t = y_{t-1} + D(y_{t-1} - y_t)$; a bill return is a lagged short rate; and deflating both by that country's own price index gives a real return that was measured rather than modelled.

Table 4. Recorded series that exist and still cannot be used

Country	Series	Source	From	To	Observed years
Austria	bond	Clio-Infra bond yields	1966	2010	45
Canada	bill	Jordà–Schularick–Taylor yields and short rates	1935	2020	86
Canada	bond	Jordà–Schularick–Taylor yields and short rates	1891	2020	130
Ireland	bill	Jordà–Schularick–Taylor yields and short rates	1923	2020	98
Ireland	bond	Jordà–Schularick–Taylor yields and short rates	1923	2020	98
New Zealand	bond	Clio-Infra bond yields	1934	2010	74

Notes. Rebuilt from published rates and deflated by each country's own price index. None of these countries has an equity return series in any source available to us, which is why none is in the panel.

None of it gets them into the panel. A lifecycle investor needs a domestic *equity* return, and the macro file carries none for these four — not even the interpolated variants it supplies elsewhere. Rates alone cannot support a portfolio choice. That is why the panel is sixteen countries and not twenty.

3.6.2 The asset class we measure and do not use

The macro file carries a fourth asset the source project measured and no result in this paper uses: **housing total returns**, empirical for all 16 observed countries over 1,805 country-years (1891–2020). It is the largest block of genuine data in our sources that nothing here reads, so we audit it rather than leave it unmentioned.

Table 5. Housing total returns: observed, and why they stay out

Country	Years	Mean real	s.d. published	s.d. de-smoothed	Equity s.d.	Autocorr.
Australia	120	6.2%	11.7%	11.7%	16.2%	-0.02
Belgium	119	7.9%	15.2%	21.8%	22.9%	0.34

Country	Years	Mean real	s.d. published	s.d. de-smoothed	Equity s.d.	Autocorr.
Switzerland	119	5.7%	6.5%	10.8%	18.9%	0.47
Germany	96	6.6%	8.5%	8.5%	33.5%	-0.13
Denmark	130	8.0%	7.8%	12.1%	17.2%	0.42
Spain	120	5.3%	11.8%	12.2%	20.8%	0.03
Finland	99	9.4%	15.2%	24.0%	30.0%	0.47
France	130	6.3%	10.3%	21.5%	23.1%	0.63
United Kingdom	118	5.4%	9.0%	12.5%	18.7%	0.33
Italy	86	4.7%	9.3%	9.3%	26.6%	-0.05
Japan	75	6.6%	8.1%	12.0%	25.8%	0.46
Netherlands	130	7.4%	9.0%	14.9%	21.2%	0.46
Norway	130	7.8%	8.6%	10.3%	20.1%	0.17
Portugal	73	6.6%	8.5%	11.5%	26.0%	0.33
Sweden	130	8.1%	8.9%	12.8%	20.3%	0.35
United States	130	6.2%	7.9%	7.9%	18.7%	-0.17

Notes. Real annual total returns on the national housing stock, from the same source as the equity series. De-smoothing is first-order Geltner using each country's own lag-one autocorrelation.

The comparison is the one the source project is known for. Median real housing returns are **6.6%** against **6.9%** for the same countries' equity — indistinguishable — at a published standard deviation of 8.9% versus 21.0% — a ratio of 0.43. Equity-like returns at that volatility would look dominant in any mean-variance comparison in this paper, which is precisely why the series deserves scrutiny rather than adoption.

A house price index is built from appraisals and sparse transactions, and that smooths it. The median lag-one autocorrelation of housing returns is 0.34 against 0.06 for equity, and housing is the more autocorrelated series in 12 of 16 countries. Undoing that to first order raises the median standard deviation from 8.9% to 12.0%: most of the apparent free lunch is a measurement artefact.

Even de-smoothed the series is not investable as written. It is an unlevered, untaxed, frictionless total return on an entire national housing stock, with no transaction costs, no vacancy, no maintenance and none of the single-property concentration a real household bears. Treating it as a fourth sleeve would overstate what a household can buy and would change the headline result. We measure it, record it, and leave it out — deliberately, and on the record.

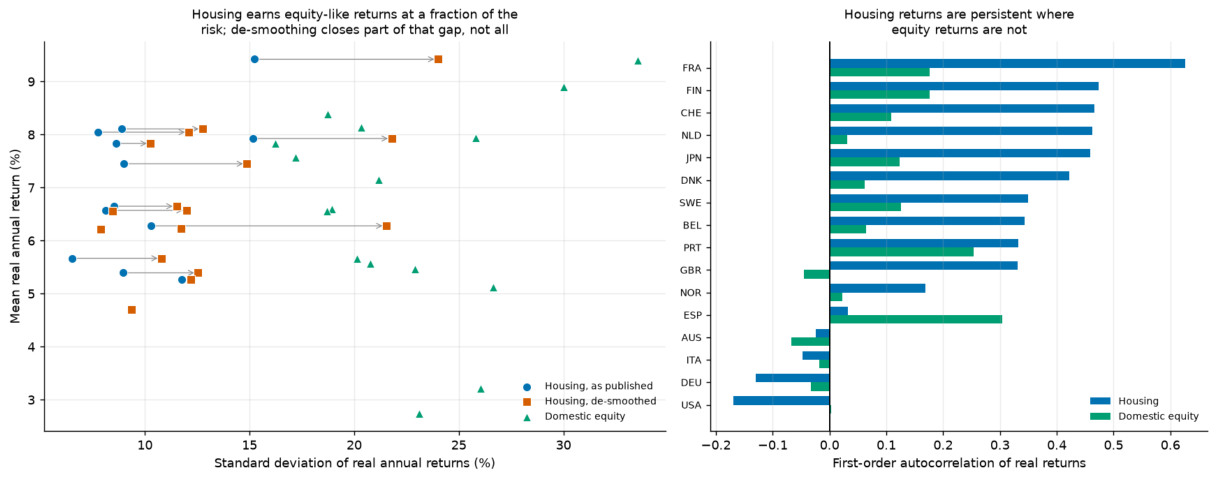

Figure 2.

Left: risk and return by country for housing as published, for housing with its own first-order smoothing undone (arrows), and for that country's domestic equity. Right: the lag-one autocorrelation that does the work. De-smoothing closes part of the volatility gap and not all of it, which is why housing is reported here rather than added to the investable set.

3.6.3 The series that bears on our income model

The same file carries a nominal wage index for all 18 of its countries, including the two with no return series. Deflated by each country's own consumer prices it measures economy-wide **real wage growth** over 2,269 country-years (1891–2020). The median country compounded real wages at **1.41% a year**, from -0.02% (Belgium) to 3.16% (Finland) — which over the 37-year career we simulate compounds to 1.68×.

Table 6. Measured real wage growth, and what our income profile assumes

Country	Years	Real wage growth p.a.	Excl. war years	s.d.	Compounded over a career
Australia	130	1.43%	1.61%	4.0%	1.72
Belgium	124	-0.02%	1.20%	17.3%	0.99
Canada	130	1.36%	1.27%	2.5%	1.67
Switzerland	130	1.62%	1.75%	3.8%	1.84
Germany	128	0.88%	1.03%	7.9%	1.39
Denmark	130	2.01%	2.05%	6.1%	2.13
Spain	122	1.98%	2.68%	10.3%	2.11
Finland	130	3.16%	3.81%	15.5%	3.26
France	130	1.11%	1.85%	5.2%	1.52
United Kingdom	130	1.29%	1.41%	4.6%	1.63
Ireland	77	2.83%	2.75%	3.2%	2.89
Italy	128	0.98%	2.26%	10.7%	1.45
Japan	130	2.37%	2.66%	13.8%	2.43
Netherlands	130	1.12%	1.18%	4.7%	1.53
Norway	130	1.83%	1.79%	3.8%	1.99
Portugal	130	0.74%	1.03%	9.9%	1.32
Sweden	130	2.43%	2.56%	4.3%	2.49
United States	130	1.40%	1.31%	2.5%	1.69

Notes. Geometric means, because they are what compound across a career. Our deterministic profile implies 1.18% a year over the same span.

The war years carry more of that spread than the economics does, which is why the table reports the series both ways. The two largest observations in the panel are Belgium 1919 (+151%) and Finland 1918 (-66%): a wage index spanning occupation, rationing, suppressed prices and post-war repricing is measuring those at least as much as it is measuring wages. Dropping 1914–1919 and 1939–1946 lifts the median to 1.77% (+0.36 percentage points) and moves the lowest country, Belgium, from -0.02% to 1.20% — from an implausible claim about a century of that country's wages to an unremarkable one. We keep them in the headline number, because a worker alive then lived through them, and note that excluding them only widens the gap we are about to describe.

Our income profile has no term for it. Section 4.3 sets real labour income as a deterministic hump peaking at age 50 at 1.76× starting income and ending the career at 1.54× — an average of 1.18% a year. That is an *age* effect: the progression a worker earns by getting older. Economy-wide wage growth is a different quantity, lifting the whole distribution regardless of age, and in the Cocco–Gomes–Maenhout estimation our profile is taken from the two are separated by construction and are therefore additive. A worker facing both would see roughly 2.61% a year.

That caveat decides the size of the gap rather than its existence: whether the components add depends on how the source profile was estimated, and one fitted to a panel that still carried time effects would already absorb part of the growth. What is not in doubt is that the quantity is measurable, that eighteen countries measure it, and that nothing in our pipeline reads it.

The direction is the one that matters for reading this paper. Understating income growth understates human capital throughout the career, and human capital is the bond-like asset in the standard lifecycle argument — so having less of it *weakens* the case for holding equity when young. The bias runs against our conclusion rather than toward it, as most of the known biases here do. The effect on the savings analysis of Sections 12 and 13 is genuinely ambiguous, because faster income growth raises both what a given savings rate accumulates and the consumption it must replace, and this audit does not resolve it. Re-estimating the income process is a modelling change rather than a data one, so we record it as a quantified limitation rather than apply it silently.

3.6.4 Is the source we kept genuine?

The workbook was obtained from a redistributed copy rather than downloaded from the compilers, so it is audited rather than assumed. Two tests, in opposite directions.

Table 7. The workbook against independently known annual returns

Observation	This workbook	Independently known	Difference	Passes
US equities in the worst year of the slump	-0.4030	-0.4300	0.0270	yes
US equities in the global financial crisis	-0.3875	-0.3700	-0.0175	yes
US equities in the 1974 bear market	-0.2544	-0.2600	0.0056	yes
US equities rebounding in 1933	0.5264	0.5400	-0.0136	yes
US equities in 2013	0.2956	0.3200	-0.0244	yes

Notes. Nominal equity total returns for years whose magnitude is not in dispute. A reconstructed or synthetic file would not reproduce both directions of the 1931-33 swing and the 2008 drawdown.

All 5 of 5 land within tolerance. The second test runs the other way. Equity total return should satisfy the accounting identity $eq\_tr = (1 + \textit{capital gain})(1 + \textit{dividend yield}) - 1$; across 2,163 observations it fails by more than one part in a million 38% of the time. **That failure rate is evidence of authenticity.** The components in the real database are sometimes spliced from different underlying indices, so they do not always reconcile; a file that satisfied the identity everywhere to machine precision would be one whose components had been back-solved from its totals.

3.6.5 One finding that does not pass

The last five years of every equity series are smoother than that country's own history. **All 16 of 16 countries** show a lower standard deviation over 2016–2020 than over 1950–2015, with a median ratio of 0.54. A five-year standard deviation is far too noisy to read one country at a time, which is why this is a sign test: under a fair-coin null the probability of all 16 pointing the same way is **3.1e-05**. Spot-checking the United States sharpens it — the workbook records +14.2% for 2018, a year in which every broad US index fell, and +8.2% for 2019 against roughly +31% for the S&P 500.

The likely explanation is that the redistributed copy was extended past the compilers' own end date by someone else. **We therefore treat 2016–2020 as unverified.** Those five years are under four percent of the panel's country-years and cannot move a 68-year lifecycle result materially, but the finding is recorded rather than left for a reader to discover.

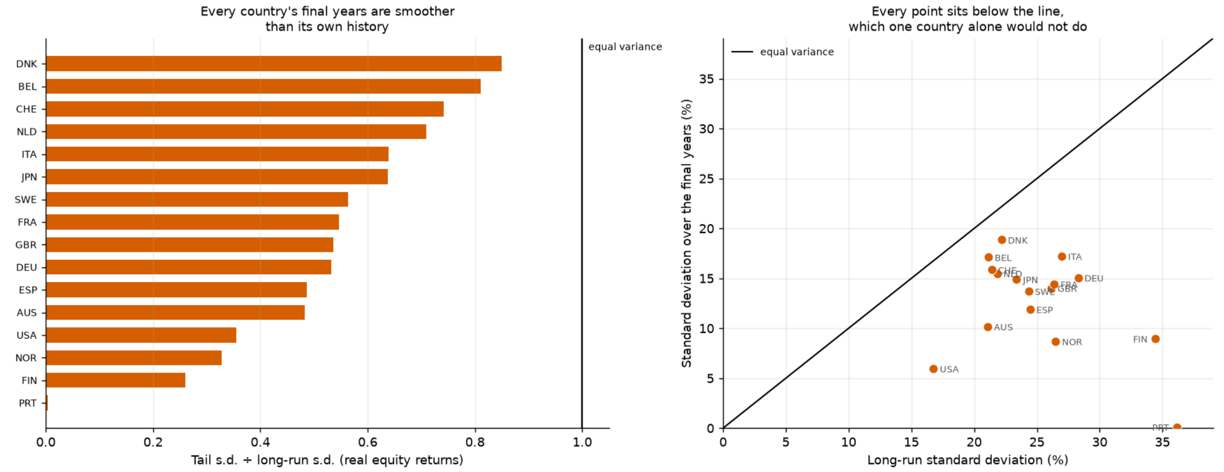

Figure 3.

The variance test, read two ways. Left: each country's standard deviation over the final five years as a ratio of its own 1950–2015 standard deviation. Right: the two standard deviations plotted against each other, with the 45-degree line. One country below the line would be unremarkable; every country below it is a property of the file rather than of the markets.

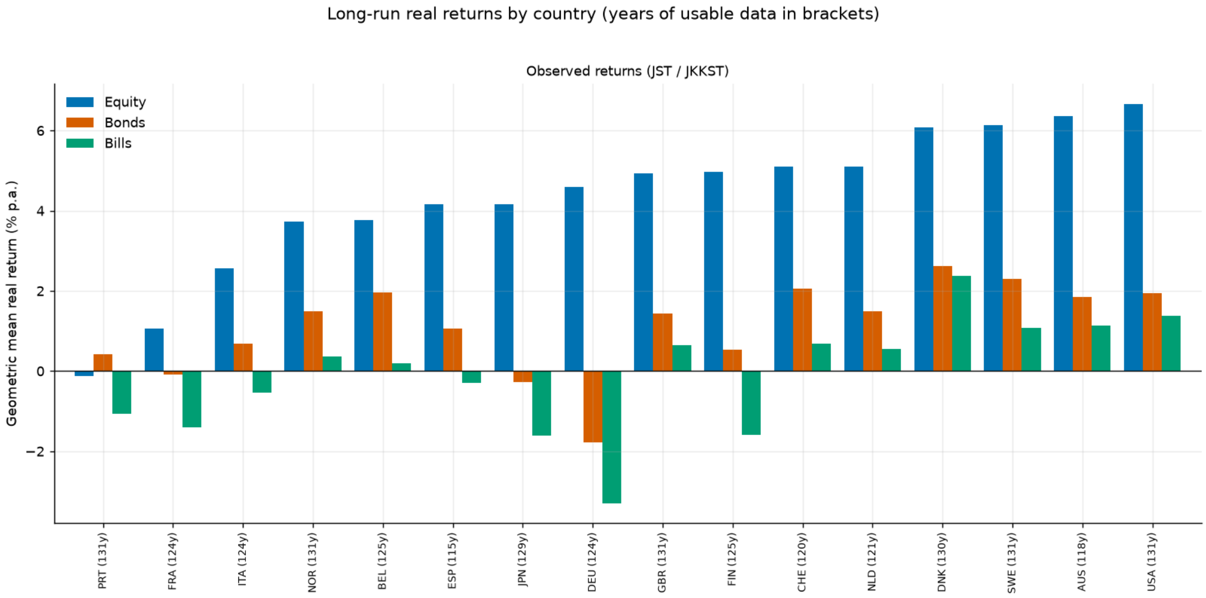

Figure 4.

Distribution of annual real returns by country and asset class. The left tails are the point of the panel: several countries record single-year real equity losses beyond −60%, and these are not outliers to be trimmed but the events a lifecycle investor is exposed to.

4. Methodology

4.1 The cross-country stationary block bootstrap

A simulated lifetime is 68 years long and is assembled from contiguous blocks of history. The sampler is a stationary block bootstrap in the sense of Politis and Romano (1994): block lengths are drawn from a geometric distribution with mean 10 years, truncated to [1, 40], and blocks are laid end to end until the horizon is filled.

Two features distinguish it from a textbook block bootstrap and both matter for the result.

- **Blocks are calendar-joint.** A block is a (country, start-year, length) triple, and the same triple is applied to all five series simultaneously. Domestic equity, international equity, bonds, bills and inflation for a given simulated year therefore come from the same real country in the same real year. This is what preserves the cross-asset correlation matrix of Table 2 without any of it being imposed parametrically.
- **Blocks respect data gaps.** Admissibility is enforced through pre-computed run-lengths: for each (country, year) the sampler knows how many consecutive years of complete data follow, and only draws a block that fits inside a run. A wartime hole is never bridged.

Across the production run the sampler drew 30,848 blocks with a mean realised length of 8.82 years against a target of 10, a median of 6, a ninetieth percentile of 20 and a maximum of 40. The realised mean falls short of the target for a mechanical reason worth stating: truncation at the maximum and the gap-admissibility constraint both bite from above, so the effective block length is always slightly below the nominal one.

The country for a lifetime is drawn once (*per_lifetime*) with probability proportional to the length of that country's usable history (*history*). Both choices are varied in Appendix C: redrawing the country at every block, and weighting countries uniformly, both leave the ranking unchanged. The weighting choice has little room to bite here, because every country carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. The choice would matter on a panel with short histories, where a country covering only one era could otherwise speak as loudly as one covering a century and a half.

Table 8. Does the bootstrap reproduce the panel it samples from?

Series	Bootstrap mean	Panel mean	Gap (bp)	Bootstrap s.d.	Panel s.d.	s.d. ratio	Bootstrap kurtosis
Domestic equity	6.86%	6.77%	8.3	22.87%	22.50%	1.016	2.9
International equity	8.52%	8.33%	19.0	22.27%	21.69%	1.027	42.5
Bonds	2.02%	1.98%	3.9	11.51%	11.81%	0.974	6.0
Bills	0.58%	0.44%	14.0	7.37%	7.99%	0.922	24.0
Inflation	5.23%	5.97%	-74.1	18.76%	37.66%	0.498	1282.5

Notes. Means agree to within twenty basis points on every series and standard deviations to within eight percent. The inflation row has an excess kurtosis in the thousands: that is the hyperinflation episodes surviving into the simulated paths, which is the intended behaviour rather than a defect.

The correlation matrix is reproduced to a maximum absolute deviation of 0.031 across the four asset-to-asset pairs that drive portfolio behaviour. The largest deviation anywhere in the matrix is 0.068 and it sits on the inflation row, where the excess kurtosis reported above makes the sample correlation itself unstable. We report both figures rather than the flattering one.

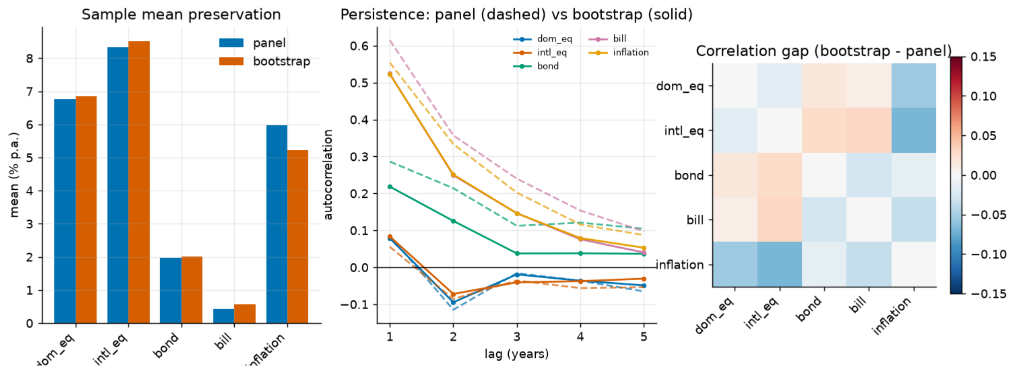

Figure 5.

Bootstrap diagnostics. Simulated marginal distributions against the panel they are drawn from, the preservation of the cross-asset correlation structure, and the within-path autocorrelation that block sampling is designed to retain.

Block length and long-horizon dispersion

Block length is the single most consequential tuning parameter in the design, because it controls how much multi-decade persistence survives into a simulated lifetime. The naive expectation is that longer blocks produce more dispersion in long-horizon outcomes. The data say the opposite, and the reason is instructive.

Table 9. Sensitivity of 68-year outcomes to the mean block length

Mean block (years)	Mean annualised	S.d. annualised	5th pct	95th pct	Within-path AR(1)
1	4.41%	3.31%	-1.52%	9.49%	-0.014
5	4.49%	3.27%	-1.41%	9.25%	0.064
10	4.43%	3.20%	-1.53%	8.89%	0.079
20	4.37%	3.14%	-1.69%	8.68%	0.088

Notes. Longer blocks raise the within-path autocorrelation, as intended, but slightly *reduce* the dispersion of annualised 68-year outcomes. Over a horizon this long the historical series mean-reverts, so preserving more of its internal structure preserves that mean reversion too.

Within-path autocorrelation rises monotonically from -0.014 at one-year blocks to 0.088 at twenty-year blocks, confirming that the sampler is doing what block sampling is for. But the standard deviation of annualised 68-year returns *falls* slightly over the same range. This is a long-horizon mean-reversion finding, not a bug: a lifetime that inherits real historical sequences inherits their tendency to revert, whereas one assembled from independent annual draws does not. It is also a caution against reading the block length as a simple dial for "more risk".

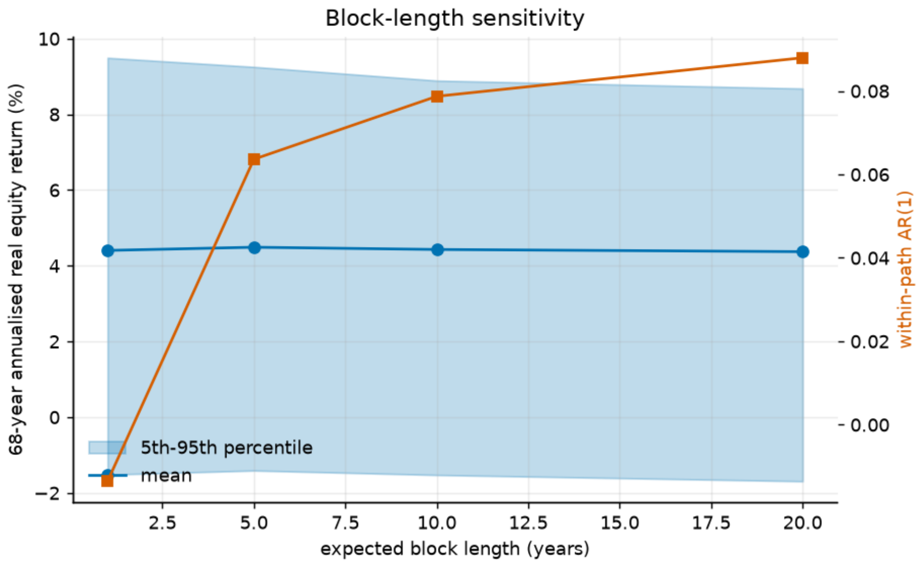

Figure 6.

Block-length sensitivity. Long-horizon dispersion is close to flat in the mean block length even as within-path persistence rises steadily, which is the signature of mean reversion in the underlying series.

4.2 The lifecycle model

A simulated investor begins work at age 25, retires at 63 and dies at 93, giving a 68-year horizon of which 38 are working years. They save a fixed fraction s = 10% of labour income, rebalance annually to the strategy's target weights, and on retirement switch to a withdrawal rule.

Wealth evolves as

$$W_{h+1} = (W_h + c_h - x_h) \cdot (1 + r^p_h), \quad r^p_h = \Sigma_a w_{a,h} r_{a,h} \quad (1)$$

where c_h is the contribution in working years and zero afterwards, x_h is the withdrawal in retirement and zero before, $w_{a,h}$ is the strategy's weight on asset a at age h, and all returns are real. Consumption is labour income net of saving while working, and the sum of the social-security benefit and the portfolio withdrawal in retirement:

$$C_h = (1 - s)\, Y_h \ (working) \quad C_h = B + x_h \ (retired) \quad (2)$$

Labour income follows a deterministic hump-shaped profile (b_1 = 0.045, b_2 = -0.0009 in age and age squared) multiplied by permanent and transitory shocks with standard deviations 0.10 and 0.25. The permanent shock is a cumulative sum, so income dispersion widens over a career; the transitory shock is i.i.d. The profile is an age effect and carries no economy-wide wage growth; Section 3.6.3 measures what that leaves out and which way it biases the result.

The social-security benefit uses the US primary-insurance-amount bend-point schedule rather than a flat replacement rate, applying 90%, 32% and 15% to successive tranches of career-average earnings. This is not decoration. A flat replacement rate scales with the outcome and therefore insures nothing; a progressive schedule supplies a genuine real consumption floor, which is exactly what determines how much the left tail of a risky strategy actually costs.

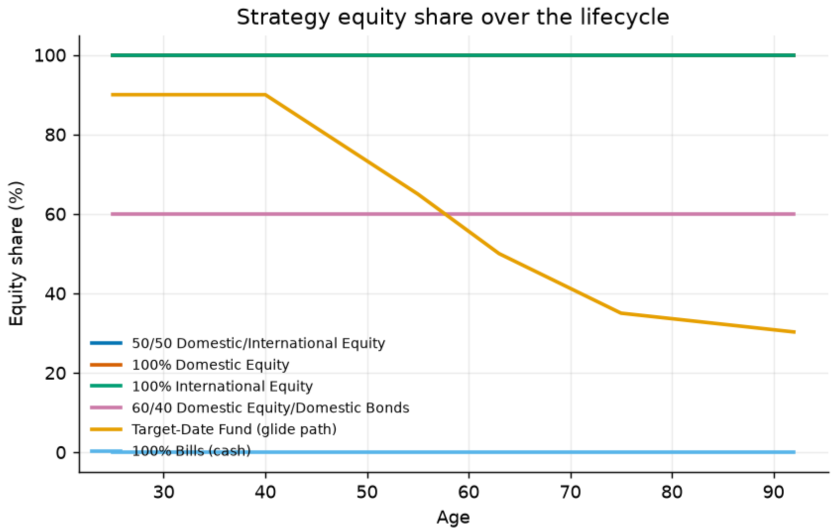

Figure 7.

The six benchmark allocation strategies as equity share by age. The target-date fund is the industry-standard declining glide path; the comparison strategies are held at fixed weights.

4.3 Preferences and the certainty equivalent

Strategies are ranked by certainty-equivalent consumption: the constant, riskless consumption stream that would leave the investor indifferent to the risky one their strategy produces. Under constant relative risk aversion with discount factor β and curvature γ, lifetime utility over the evaluation window is

$$V = \Sigma_h \beta^h u(C_h) + \beta^H \theta \cdot u(\kappa + W_H), \quad u(c) = c^{1-\gamma} / (1 - \gamma) \quad (3)$$

and the certainty equivalent inverts that utility back into consumption units. We report risk aversions of {2, 5, 10} with γ = 5 as the baseline, β = 0.96, and a bequest weight θ = 2.

The bequest term uses the shifted-power specification of De Nardi (2004) with κ = 1. The shift is not cosmetic. Without it, any path that dies with exactly zero wealth contributes $u(0) = -\infty$ at $\gamma \geq 1$, one such path in a hundred thousand collapses the entire certainty equivalent, and the ranking becomes a statement about which strategy avoids exact zeros rather than about which delivers better consumption. This is a genuine trap in lifecycle work and we flag it because we walked into it.

We also report Epstein–Zin preferences in their ex-ante, early-resolution form, which separates risk aversion from the elasticity of intertemporal substitution, with ψ taking the values {0.5, 1.5}. Under CRRA these two are locked together at ψ = 1/γ, and a reader is entitled to ask whether the result is being driven by the intertemporal channel rather than the risk channel. Section 6.2 shows it is not.

One further specification choice deserves emphasis. Utility is evaluated over the **retirement** window rather than the whole lifetime for the allocation comparisons. The reason is mechanical: with a fixed savings rate, working-life consumption is *identical* across allocation strategies by construction, so including it adds a large constant to every strategy's utility and compresses the differences that the exercise is about. Where a policy *does* change working-life consumption — the retirement timing and savings-rate studies of Sections 12 to 14 — we switch to the whole-lifetime window and say so explicitly.

4.4 The comparison discipline

Most of the extensions in this paper compare policies that differ in more than one way at once, and the arithmetic of a certainty equivalent will happily credit the wrong difference. Four rules are applied throughout.

- **Matched baselines.** A rule that retires people later is scored against a fixed date matched on the same *mean* retirement age, not against the original age 63. A savings rule that drifts to saving more is scored against a constant rate interpolated at its own realised career average. Without this, a policy is rewarded for working longer or saving more rather than for being smarter.
- **Common random numbers.** Every sweep and every optimiser reuses the same bootstrap draws and the same income shocks, so a difference between two settings is a policy effect and not a sampling artefact. This also makes the objective a deterministic function of the policy, which is what licenses coordinate ascent.
- **Deviation profiles.** Before any structure in a solved schedule is described in words, each element is reset to a neutral reference and the cost measured in basis points. A solved glide path can look highly structured while most of its structure is worth nothing; the profile separates the two.
- **Grid-edge checks.** An optimum sitting on the boundary of its own search grid is a truncation, not an optimum. Every such case is flagged in the text and, where it mattered, the grid was widened and the analysis rerun.

These are not stylistic preferences. Section 12 reports a result that fell by roughly sixty percent when a matched baseline replaced an unmatched one, Section 8 reports apparent glide-path structure that the deviation profile dissolved, and Section 14 reports a functional-form ranking that reversed its interpretation once the grid-edge check was applied.

4.5 Implementation and verification

The model is implemented in Python with NumPy and pandas. Portfolio returns for a whole cohort are formed as a batched matrix product and the wealth recursion runs vectorised across paths, which is what makes a hundred thousand lifetimes per strategy — and the tens of thousands of policy evaluations the optimisers require — tractable on a single core.

Correctness is enforced by 523 automated tests. The load-bearing ones are equivalence tests: the path-dependent engine used for flexible retirement and conditional saving must reproduce the fixed-date engine *bit for bit* when given a fixed date and a constant rate, and the batched glide-path evaluator must reproduce the reference simulator to stated floating-point tolerance. Every extension in this paper is built on a simulator that is asserted equal to the one that produced the baseline.

Appendix D sets out how the computation is organised, what the test suite establishes, and the discipline by which the prose in this paper is held to the tables it describes.

5. Baseline Results

All results in this section use 100,000 simulated lifetimes per strategy, drawn from the same bootstrap sample so that strategies face identical market histories. Certainty equivalents are over the retirement window, for the reason given in Section 4.3.

5.1 The headline comparison

Table 10. Headline lifecycle outcomes by strategy

Strategy	CEC γ=2	CEC γ=5	CEC γ=10	P(ruin)	Median wealth at 63	Median cons.	5th pct cons.	Median bequest
50/50 domestic/international equity	1.406	1.048	0.756	11.7%	21.2	1.483	0.707	44.7
100% domestic equity	1.183	0.882	0.668	25.4%	16.1	1.241	0.579	16.7
100% international equity	1.521	1.108	0.782	9.0%	24.4	1.613	0.755	64.2
60/40 domestic equity/bonds	1.105	0.871	0.671	24.2%	13.3	1.149	0.585	10.2
Target-date fund (glide path)	1.214	0.943	0.704	22.2%	16.4	1.273	0.630	9.4
100% bills (cash)	0.847	0.739	0.618	56.4%	6.4	0.856	0.521	0.0

Notes. Consumption and wealth are expressed as multiples of the investor's real income at age 25. P(ruin) is the probability of exhausting the portfolio with retirement years still to fund. 100,000 paths per strategy on common bootstrap draws.

The ordering is unambiguous at every risk aversion tested. At γ = 5 the 50/50 all-equity portfolio delivers a certainty equivalent of 1.048 against 0.943 for the target-date fund, an advantage of 11.2%. Against the 60/40 portfolio the advantage is 20.3%, and against bills 41.8%.

What makes the result more than a restatement of the equity premium is the tail behaviour. The all-equity portfolio has a *lower* ruin probability (11.7%) than the target-date fund (22.2%), a higher fifth-percentile retirement consumption (0.707 against 0.630), and a lower probability of falling below a seventy-percent replacement target (13.6% against 22.6%). The conservative portfolio is not buying downside protection on this panel. It is paying for the appearance of it.

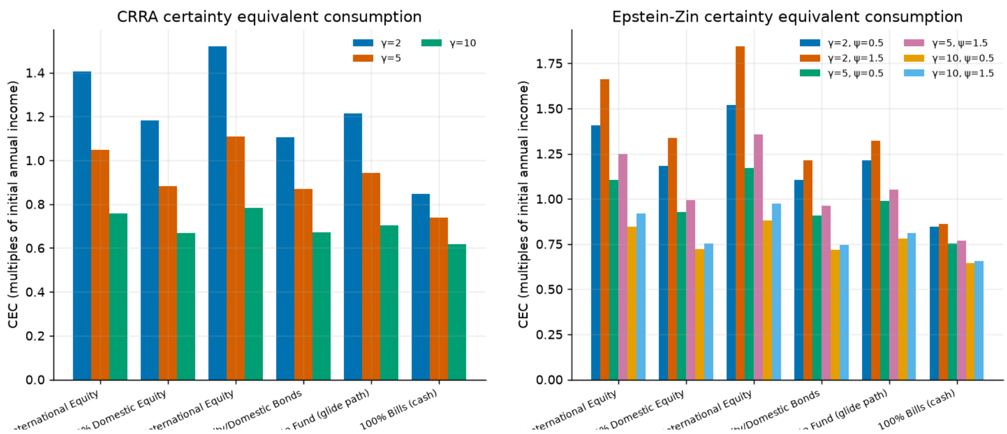

Figure 8.

Certainty-equivalent consumption by strategy and risk aversion. The ranking is stable in γ; the gap narrows as risk aversion rises but does not close within the range tested.

5.2 The mechanism is international, not equity

Domestic equity alone is *not* the winning strategy. Its certainty equivalent at γ = 5 is 0.882, below the 50/50 portfolio's 1.048, and its ruin probability is 25.4% — nearly double the diversified portfolio's. Pure international equity does better still (1.108), which is a consequence of the leave-one-out construction rather than a recommendation: an investor holding "international" equity in this model holds a 37-country average, and no single real investor has that opportunity set without also holding their own market.

The practical reading is that the case for equities here is a case for *diversified* equities. A single national market can and did deliver multi-decade real losses; the cross-section of national markets did not do so simultaneously. This is a claim about the covariance structure of the panel, and it is the reason the exercise needs an international sample rather than a longer US one.

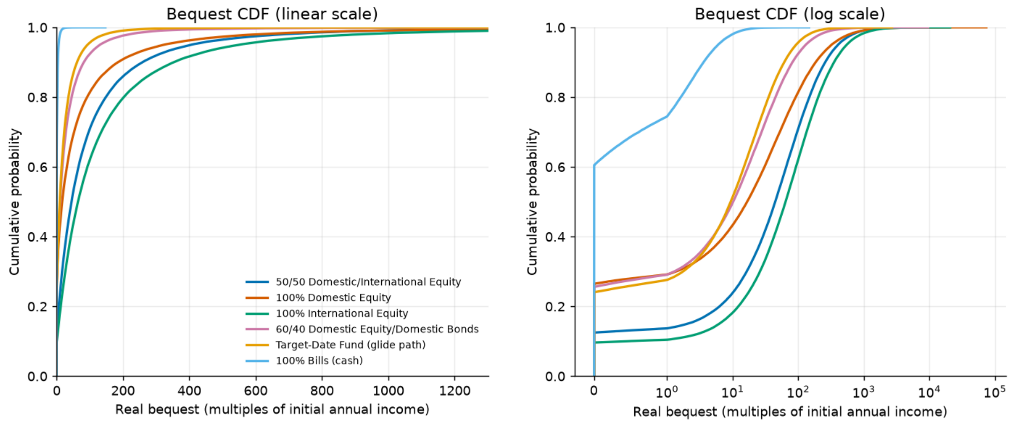

Figure 9.

Cumulative distribution of wealth at retirement. The all-equity distributions stochastically dominate the conservative ones over almost the entire range, including the left tail where the conservative case is usually made.

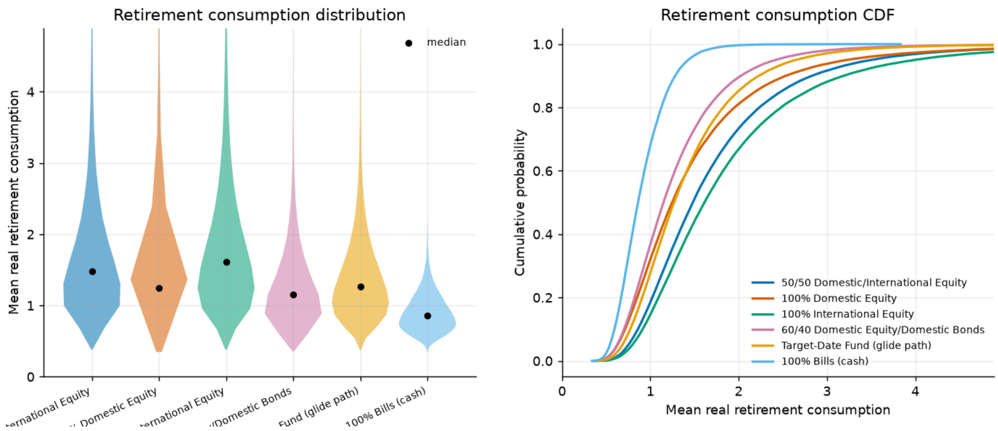

Figure 10.

Distribution of average real retirement consumption by strategy. The vertical line marks the seventy-percent replacement target used in the shortfall statistics.

5.3 Distributional dominance

A certainty equivalent is a single scalar and can hide a strategy that wins on average while losing where it matters. We therefore test the all-equity portfolio against each rival across 21 separate distributional criteria — percentiles of wealth at retirement, of retirement consumption and of bequest, plus ruin and shortfall probabilities — counting a tie as neither a win nor a loss.

Table 11. Distributional dominance of the 50/50 all-equity portfolio

Beaten strategy	Criteria	Won	Tied	Lost	Strict dominance
Target-date fund (glide path)	21	20	1	0	yes
60/40 domestic equity/bonds	21	20	1	0	yes
100% domestic equity	21	20	1	0	yes
100% bills (cash)	21	20	1	0	yes

Notes. A tie is counted separately rather than as a loss, since counting it either way would misstate the comparison. The single tie in each row is the zero-bequest floor, which several strategies reach.

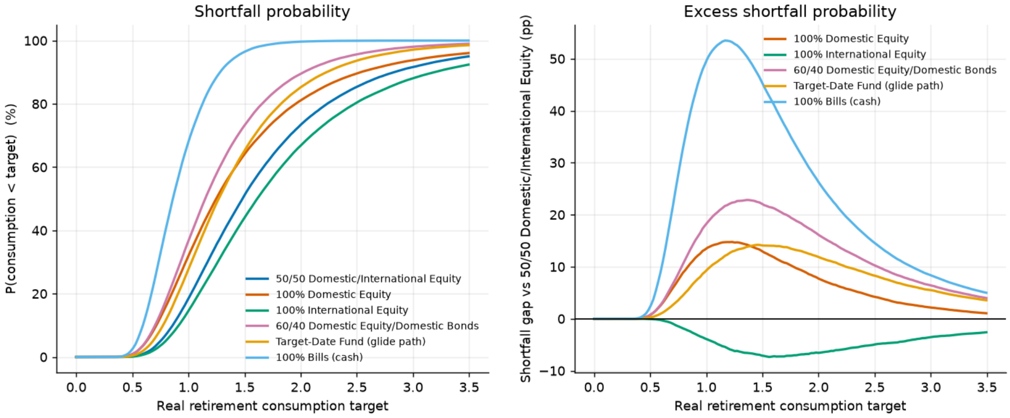

Figure 11.

Shortfall probabilities against a fixed consumption target. Measuring each strategy against a common target rather than against its own median is essential: a strategy-specific target rewards strategies with low medians.

5.4 Sustainable withdrawal rates

The four-percent rule is the most widely used number in retirement planning. It comes from US historical sequences and it does not survive this panel.

Table 12. Sustainable withdrawal rates on the international panel

Strategy	Sustainable rate at 5% ruin	Ruin probability at 4%
100% international equity	3.48%	9.1%
50/50 domestic/international equity	3.17%	11.7%
Target-date fund (glide path)	2.10%	22.0%
60/40 domestic equity/bonds	1.69%	24.0%
100% domestic equity	1.57%	25.2%
100% bills (cash)	1.07%	56.0%

Notes. The sustainable rate is the highest initial real withdrawal rate, inflation-adjusted thereafter, that leaves at most a five-percent probability of exhausting the portfolio over a 30-year retirement.

At a five-percent ruin tolerance the sustainable rate is 3.17% for the all-equity portfolio and 2.10% for the target-date fund. Withdrawing four percent produces a ruin probability of 11.7% and 22.0% respectively. Two things follow. First, the ordering of strategies is the same here as everywhere else in the paper — the equity portfolio sustains a higher rate, not a lower one. Second, the level is a serious warning: on a sample that includes the twentieth century as most countries lived it, four percent is roughly a one-in-seven proposition even on the best strategy tested.

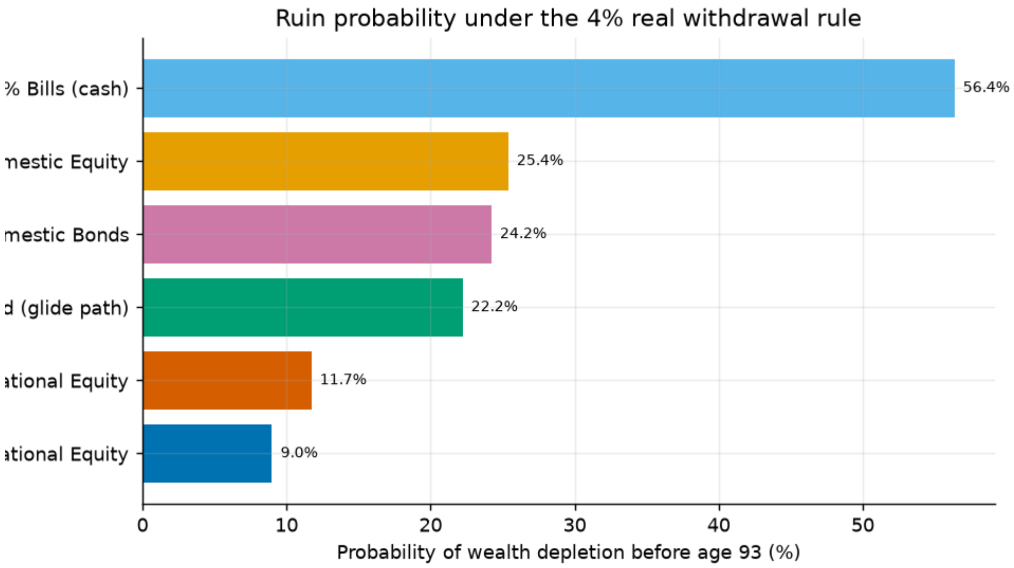

Figure 12.

Probability of portfolio exhaustion as a function of the initial withdrawal rate. The horizontal line is the five-percent tolerance used to define the sustainable rate.

5.5 How the countries are drawn

Two choices in the sampler decide how the cross-section enters a simulated life, and neither is obviously right. A lifetime's domestic country can be drawn once and held, or redrawn at every block; and countries can be weighted by the length of their recorded history or treated as equally likely. Appendix C reports both variants in full. Neither reverses a ranking.

The weighting choice has little room to bite, because every country in the panel carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. It would matter on a panel containing short histories, where a country covering a single era could otherwise speak as loudly as one covering a century and a half.

Redrawing the country at every block is the more consequential variant, and it is the design the original study uses. It implicitly assumes an investor can be resident in a different country every decade, which is not a description of anybody, but it pools the cross-section more aggressively and so gives the international mechanism its most favourable reading. Holding the country fixed for a lifetime — the specification behind every headline number here — is the more conservative of the two.

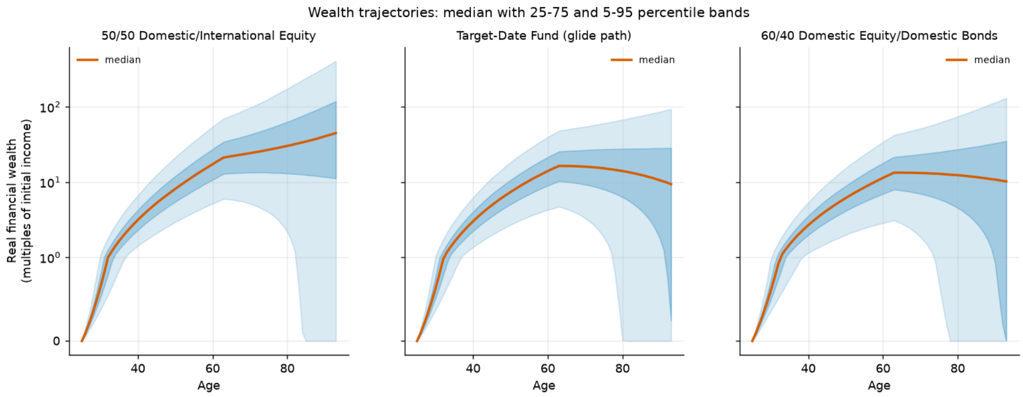

Figure 13.

Wealth trajectories by percentile for the all-equity portfolio and the target-date fund. The fan makes visible what the scalar comparison cannot: the conservative path is not narrower at the bottom, it is lower throughout.

6. Sensitivity Analysis

The headline is one point in a large parameter space. This section sweeps that space: 132 settings across ten dimensions — equity share, domestic share, risk aversion, elasticity of intertemporal substitution, bequest weight, longevity, retirement age, savings rate, withdrawal rate, social-security design, block length and return panel. Every sweep uses common random numbers, so differences between settings are parameter effects rather than sampling noise.

The ranking reverses in 0 of them. That is the single most useful sentence in this section, and the tornado below reports the range of the advantage rather than its point estimate so the reader can see how much room there is.

6.1 Tornado analysis

Table 13. Tornado: range of the all-equity advantage by parameter dimension

Dimension	Compared against	Settings	Min adv.	Median adv.	Max adv.	Range (pp)	Reversals
Risk aversion	60/40	12	5.4	19.3	33.8	28.4	0
Risk aversion	Target-date fund	12	3.5	10.5	22.2	18.7	0
Elasticity of substitution	60/40	9	20.2	25.3	34.2	14.0	0
Retirement age	60/40	6	15.7	21.4	28.7	13.0	0
Savings rate	60/40	5	14.0	20.2	26.5	12.5	0
Elasticity of substitution	Target-date fund	9	11.0	14.4	22.8	11.8	0
Withdrawal rate	60/40	13	9.0	16.3	20.2	11.3	0
Retirement age	Target-date fund	6	7.9	11.8	17.1	9.2	0
Social security design	60/40	4	20.2	24.6	29.3	9.0	0
Social security design	Target-date fund	4	11.0	13.5	18.2	7.2	0
Savings rate	Target-date fund	5	7.8	11.0	14.8	7.0	0
Bootstrap block length	60/40	6	16.2	18.9	22.9	6.7	0

Notes. Advantage is the percentage certainty-equivalent lead of the 50/50 all-equity portfolio over the named incumbent, at $\gamma = 5$ unless the dimension is risk aversion itself. Ordered by the width of the range, so the dimensions the result is most sensitive to appear first.

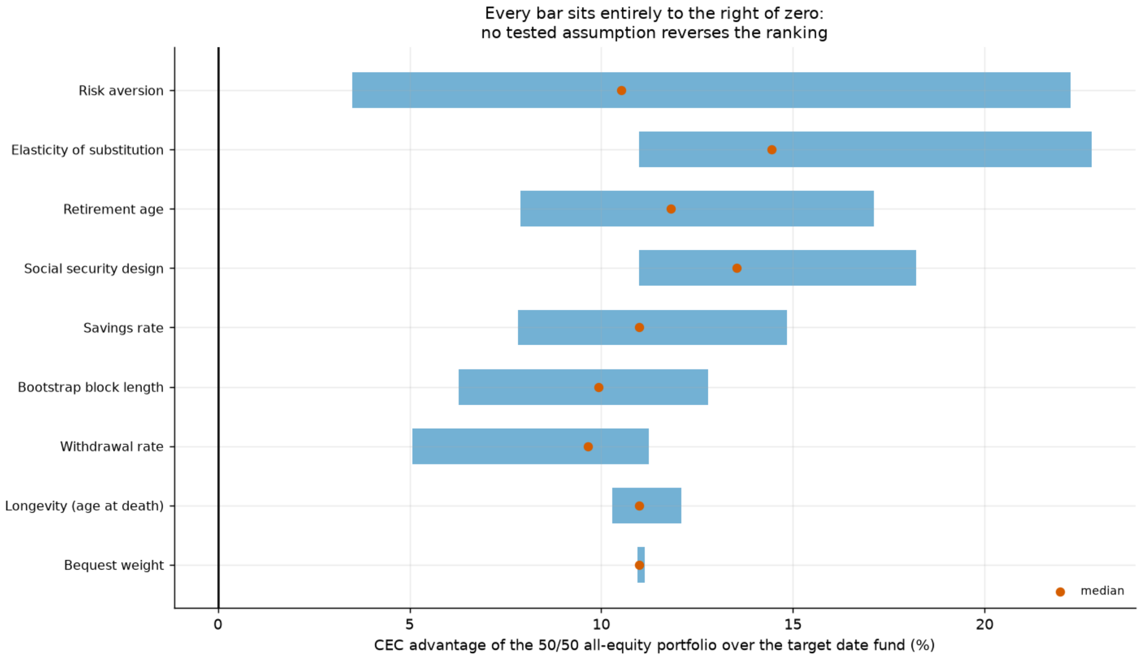

Figure 14.

Tornado plot of the all-equity advantage. The bar spans the minimum and maximum advantage observed across the settings in that dimension; no bar crosses zero.

Risk aversion is the widest dimension, as it should be: the advantage over a 60/40 portfolio ranges from 5.4% to 33.8% across γ over [1, 20]. It narrows as γ rises but does not close. Fitting the crossover point directly, the certainty-equivalent lines do not intersect within the tested range for either the target-date fund or the 60/40 portfolio; at the highest γ tested the all-equity lead is still 3.5% and 5.4% respectively.

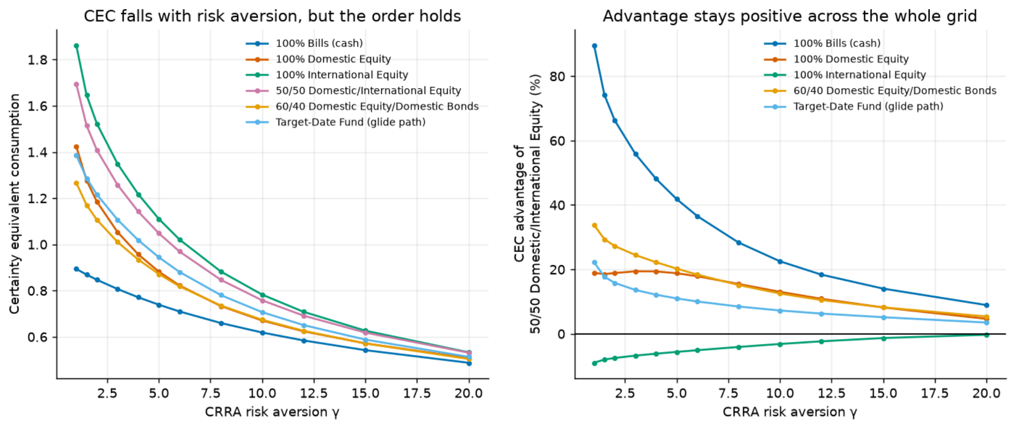

Figure 15.

Certainty equivalent by strategy across risk aversion. The conservative strategies converge toward the equity portfolios as γ rises but do not overtake them within the tested range.

6.2 Allocation sweeps and the corner solution

Sweeping the equity share continuously rather than testing discrete strategies asks a sharper question: is the optimum interior?

Table 14. Optimal equity share by risk aversion

Risk aversion	Optimal equity share	CEC at optimum	P(ruin) at optimum
γ = 2	100%	1.408	11.7%
γ = 5	100%	1.048	11.7%
γ = 10	100%	0.758	11.7%

Notes. The optimum is at the corner for every risk aversion tested. Section 8 asks the same question of the full age-by-asset schedule and reaches the same answer.

Table 15. Optimal domestic share within the equity allocation

Risk aversion	Optimal domestic share	CEC at optimum	P(ruin) at optimum
γ = 2	0%	1.522	9.1%
γ = 5	10%	1.119	8.3%
γ = 10	20%	0.793	8.1%

Notes. The home-bias question, asked of the model rather than assumed. The optimum is well below the 50% used in the headline strategy and rises with risk aversion — an artefact of the leave-one-out international construction, which gives the international leg a diversification advantage no individual investor can replicate.

The equity optimum sits at 100% at every risk aversion tested. The domestic-share optimum is more interesting: it is 0% at γ = 2 and rises to 20% at γ = 10. We do not read this as a recommendation to hold almost no domestic equity. The international leg in this model is a 37-country leave-one-out average, which is better diversified than any tradeable international index; the honest reading is that the model prefers *more* diversification than the 50/50 headline strategy provides, not that a specific number is optimal.

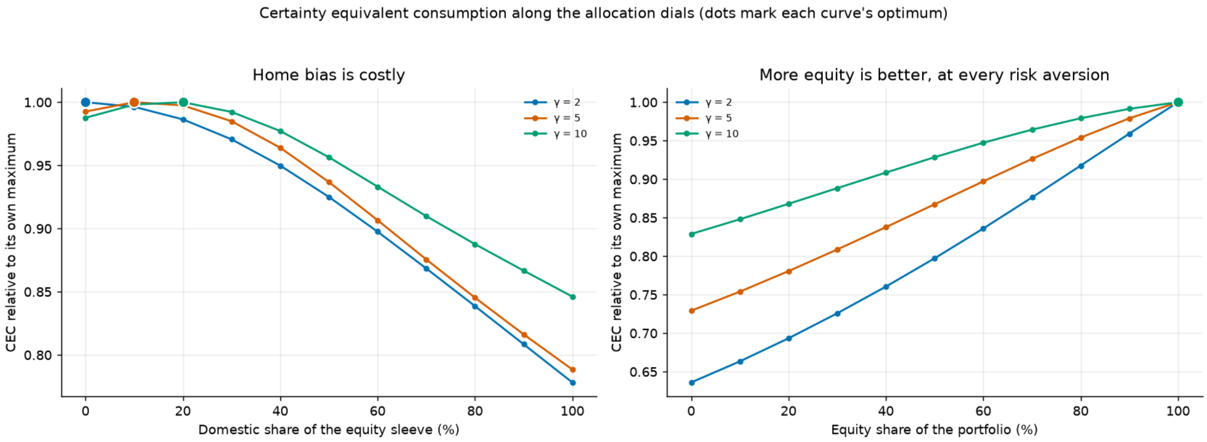

Figure 16.

Certainty equivalent across the equity share and the domestic share. The equity dimension is monotone to the corner; the domestic dimension has an interior optimum well below the 50% headline weight.

6.3 Separating risk aversion from intertemporal substitution

Under CRRA, γ and the elasticity of intertemporal substitution are the same parameter inverted, so a sceptic can reasonably ask whether the result is about attitudes to risk or about willingness to substitute consumption over time. Epstein–Zin preferences separate them. Holding γ at 5 and sweeping ψ from 0.2 to 2.5, the advantage over the 60/40 portfolio ranges from 20.2% to 34.2% and never reverses. The result is a risk result, not a substitution result.

6.4 Planning parameters

Retirement age, longevity, savings rate, withdrawal rate and social-security design are the levers an individual actually controls, and none of them changes the ranking. They do change the *size* of the advantage substantially — the withdrawal-rate dimension alone spans several percentage points — which is worth keeping in view: the all-equity lead is not a constant of nature but a quantity that depends on how the rest of the plan is set up.

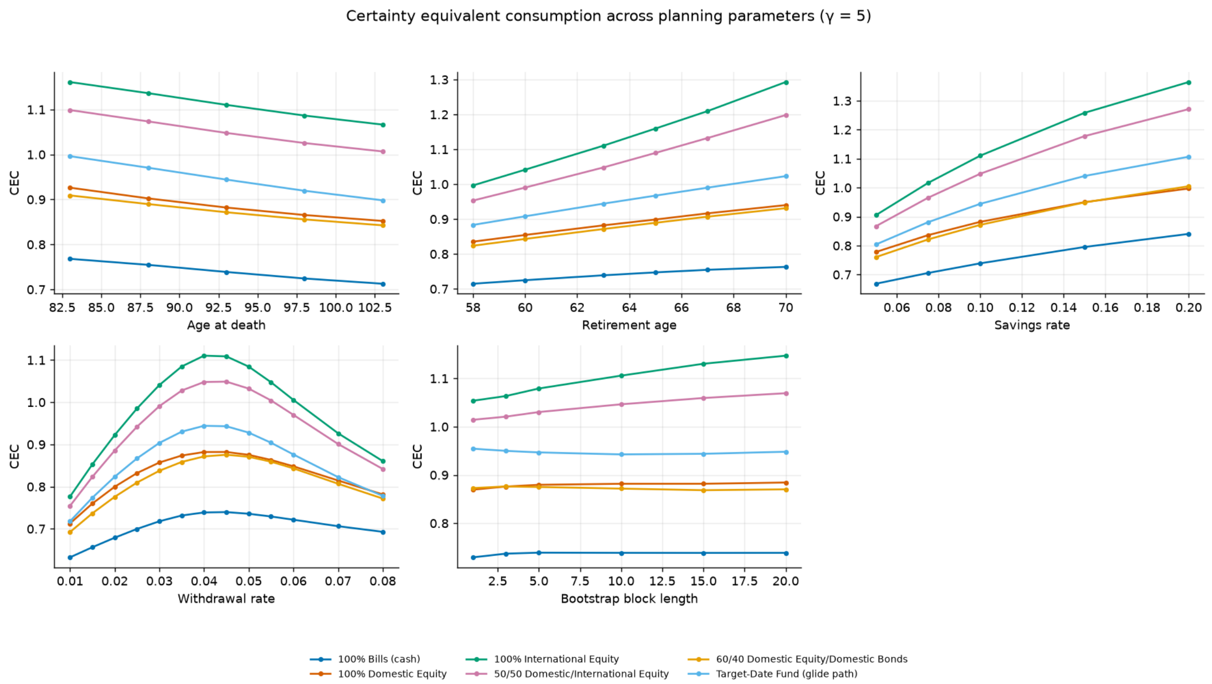

Figure 17.

Planning-parameter sweeps: retirement age, longevity, savings rate and social-security design. Ranking is preserved throughout; the magnitude of the advantage is not.

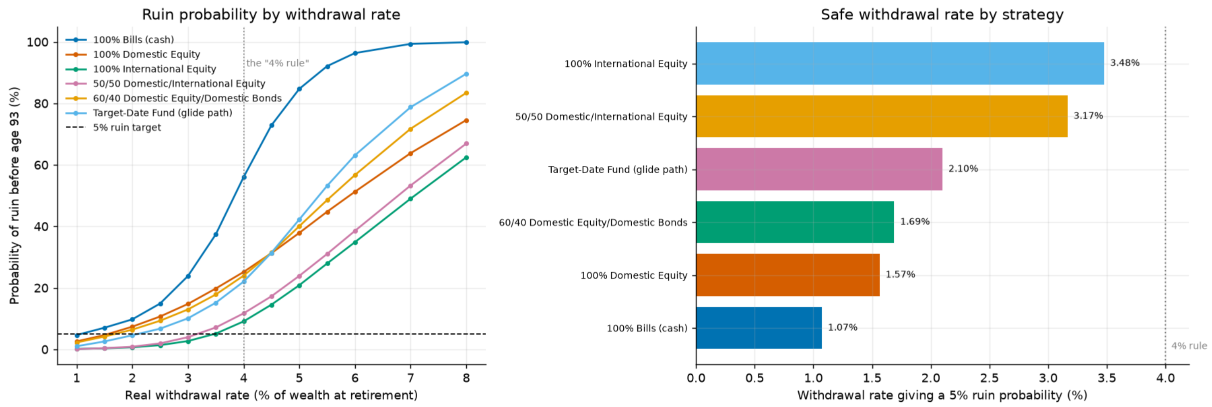

Figure 18.

The advantage as a function of the withdrawal rate. Higher withdrawal rates widen the gap, because a portfolio being drawn down faster is more exposed to the left tail the conservative strategies fail to protect against.

7. Retirement Spending Rules

The baseline draws down the portfolio with a constant real withdrawal — the four-percent rule and its variants. That is one policy among many, and it is a poor one: it ignores the portfolio entirely after the first year. This section compares 8 spending policies drawn from 8 families, each optimised over its own rate so that no family is handicapped by a badly chosen parameter.

- **Constant real.** Fixed initial percentage, inflation-adjusted thereafter. The status quo.
- **Percentage of portfolio.** A fixed fraction of the current balance each year — never runs out, but consumption is as volatile as the portfolio.
- **Guardrails.** Guyton–Klinger style: hold the real withdrawal constant until the implied rate drifts outside a band, then cut or raise it by a fixed step.
- **Floor and ceiling.** A percentage-of-portfolio rule bounded above and below relative to last year's withdrawal.
- **Endowment smoothing.** A weighted average of last year's spending and a target percentage of the current balance.
- **Required minimum distribution.** Balance divided by a regulatory life-expectancy divisor.
- **Amortisation.** The annuity payment that would exhaust the balance over the remaining horizon at an assumed return.
- **Actuarial.** As amortisation but using Gompertz life expectancy recomputed each year rather than a fixed horizon.

Table 16. Best spending rule of each family, at γ = 5

Best of family	CEC	CEC (no bequest)	P(ruin)	Median cons.	5th pct cons.	Cons. volatility	Median worst cut	Median bequest
Amortisation (annuity payment) (4% assumed return)	1.257	1.274	0.0%	2.196	0.865	0.253	40%	0.0
Actuarial (Gompertz life expectancy)	1.256	1.246	0.0%	2.132	0.846	0.265	45%	4.4
Constant % of portfolio	1.205	1.196	0.0%	1.903	0.788	0.223	48%	14.2
Endowment smoothing (light smoothing)	1.205	1.195	0.0%	1.902	0.786	0.219	45%	13.8
Guyton-Klinger guardrails (tight band)	1.200	1.192	4.1%	1.945	0.792	0.200	34%	11.3
Life expectancy / RMD	1.192	1.203	0.0%	2.242	0.849	0.427	30%	0.0
Vanguard dynamic (ceiling/floor) (asymmetric)	1.133	1.126	8.2%	1.746	0.755	0.137	18%	25.7
Constant real (4%-rule style)	1.049	1.042	17.3%	1.567	0.691	0.070	0%	35.5

Notes. Each family is evaluated at its own optimal rate. "Median worst cut" is the largest single-year real spending reduction the median path experiences — the discomfort the rule imposes in exchange for its ruin protection.

Amortisation (annuity payment) (4% assumed return) ranks first at γ = 5, with a certainty equivalent of 1.257 and a ruin probability of 0.0%. The adaptive families as a group dominate the constant-real rule, and the mechanism is not subtle: a rule that never looks at the portfolio cannot avoid exhausting it, and the certainty equivalent is extremely sensitive to the paths where it does.

The cost of that protection is spending volatility, and the table reports it explicitly rather than burying it. The best-performing rules ask the median retiree to absorb a real spending cut of tens of percent at some point in retirement. Whether that trade is acceptable is a question about the retiree, not about the model; the certainty equivalent prices it under one particular utility function and a reader with a different one should read the volatility and worst-cut columns instead.

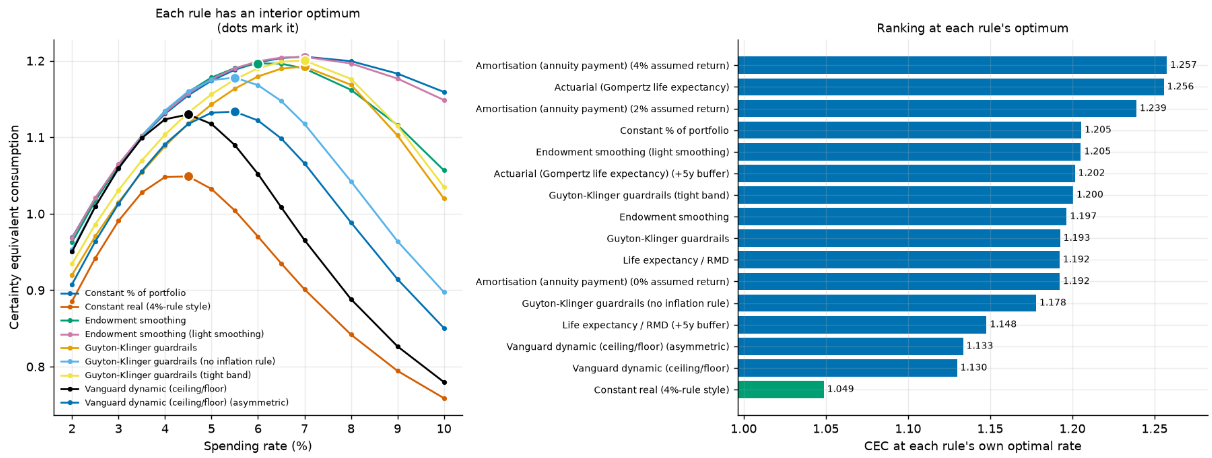

Figure 19.

Certainty equivalent as a function of the withdrawal rate for each spending family. Each family is compared at the peak of its own curve rather than at a common rate.

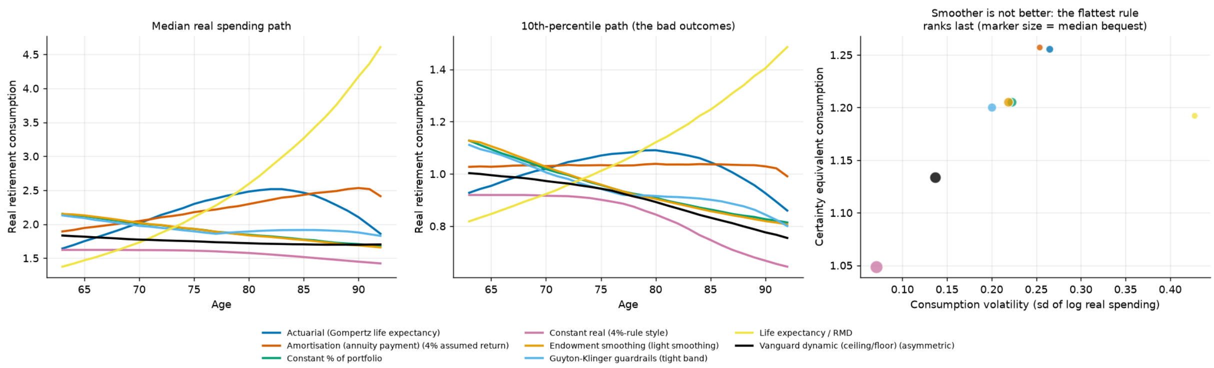

Figure 20.

Realised spending paths under each rule for a common set of market histories. The adaptive rules trade a smooth path for a solvent one.

7.1 The bequest pivot

The ranking is not invariant to the bequest motive, and this is the most interesting finding in the section. With the bequest weight switched off, the best rule is **Amortisation (annuity payment) (4% assumed return)**; with it switched on at $\theta = 2$, the ranking shifts toward rules that leave something behind. Rules that deliberately exhaust the portfolio — amortisation, actuarial — maximise consumption precisely by planning to leave nothing, which is optimal if and only if nothing is what the investor wants to leave.

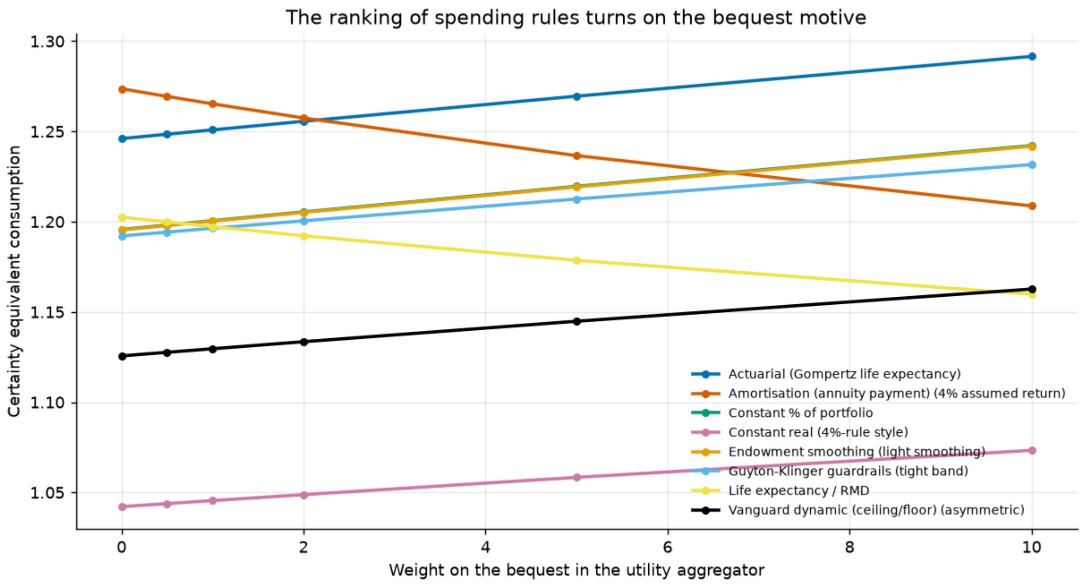

Figure 21.

Certainty equivalent by spending rule across the bequest weight. The lines cross: a rule that is optimal for an investor with no bequest motive is not optimal for one who has a strong one.

Across the 6 bequest weights tested the identity of the best rule changes, which means the question "what is the best withdrawal policy?" is not well posed without stating a bequest motive. That is a modelling result rather than a market result, but it is one that a great deal of withdrawal-rate advice ignores.

8. Solving for the Optimal Glide Path

Sections 5 and 6 compare a handful of candidate allocation schedules. That is a test, not an answer. This section solves the problem directly: what equity share should the investor hold at each of the 68 ages, if the schedule is free to be anything at all?

The objective is the certainty equivalent under common random numbers, which makes it a deterministic function of the schedule. That licenses coordinate ascent — sweeping one age at a time over a grid of weights, holding the rest fixed — because each per-age search is then exact and each full sweep is monotone. To guard against local optima the search is restarted from several initial schedules, and a relative improvement threshold is imposed so that the reported solution does not encode year-to-year jitter worth a fraction of a basis point.

Table 17. Solved schedules against the benchmark strategies at y = 5

Schedule	CEC	Gap to best (%)
free form optimal	1.117	0.00
parametric optimal	1.115	-0.20
100% international equity	1.106	-0.98
50/50 domestic/international equity	1.045	-6.46
Target-date fund (glide path)	0.941	-15.76
100% domestic equity	0.881	-21.16
60/40 domestic equity/bonds	0.870	-22.10
100% bills (cash)	0.738	-33.93

Notes. "Free-form optimal" solves an unconstrained weight at every age; "parametric optimal" fits a three-parameter logistic glide. The benchmarks are the same six strategies used throughout.

The free-form solution beats every benchmark, but by a margin that matters less than its *shape*. Across the three restarts the solved schedule holds a mean equity share of 99.4% and sits at a full equity allocation in 96% of ages. The restarts agree to within 0.012% of each other, so this is not a local optimum artefact.

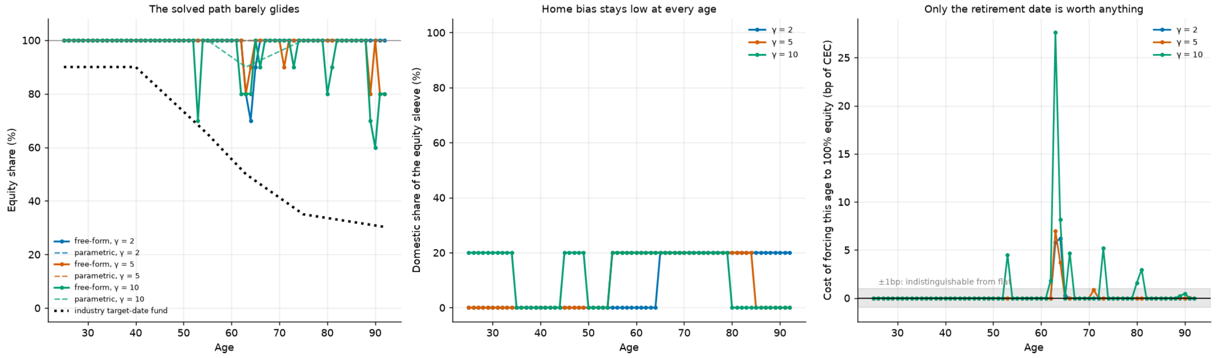

Figure 22.

The solved equity share by age, from three restarts, against the target-date fund's prescribed glide. The solution is flat at or near the corner; it is not a glide path.

8.1 How much of the solved structure is real?

A solved schedule always looks structured. The deviation profile tests whether it is: each age's solved weight is reset to a neutral reference and the certainty-equivalent cost measured in basis points. Of the 68 ages at γ = 5, only 2 move the objective by more than a single basis point. The remaining 66 are on a flat part of the surface, and any narrative built on their apparent pattern would be a narrative about search noise.

This is the reason the paper does not describe the solved glide path as having interesting age structure. It does not. The honest summary is that the optimiser goes to the equity corner and stays there, and that the small departures it makes near the ends of the horizon are worth almost nothing.

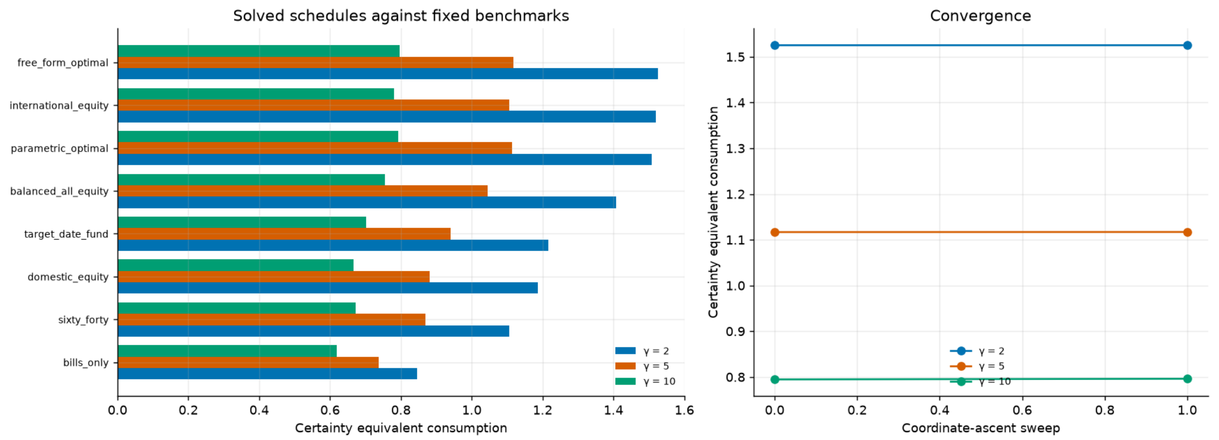

Figure 23.

Certainty equivalent of the solved schedules against the benchmarks, at each risk aversion. The solved advantage over a fixed all-equity portfolio is small; the advantage over the glide path is not.

8.2 Anchoring the glide to the retirement date

A natural objection is that the search is unconstrained and real target-date funds are anchored to a retirement date. We therefore also solve the schedule subject to an anchor: the equity share at retirement is fixed at a series of levels and the rest of the schedule is re-optimised around it.

Table 18. Solved schedules under an anchored equity share at retirement

Rule	Min Equity Share At Retirement	Mean Equity Share Elsewhere	Dip Size Pp	Solved Cec
4% rule (anchors on wealth at 63)	0.800	0.989	18.906	1.117
Constant % of portfolio	1.000	0.997	-0.313	1.314
Life expectancy / RMD	1.000	0.997	-0.313	1.277

Notes. Each row fixes the equity share at the retirement date and re-solves the remaining ages. The cost of the anchor is the certainty-equivalent gap to the unconstrained solution.

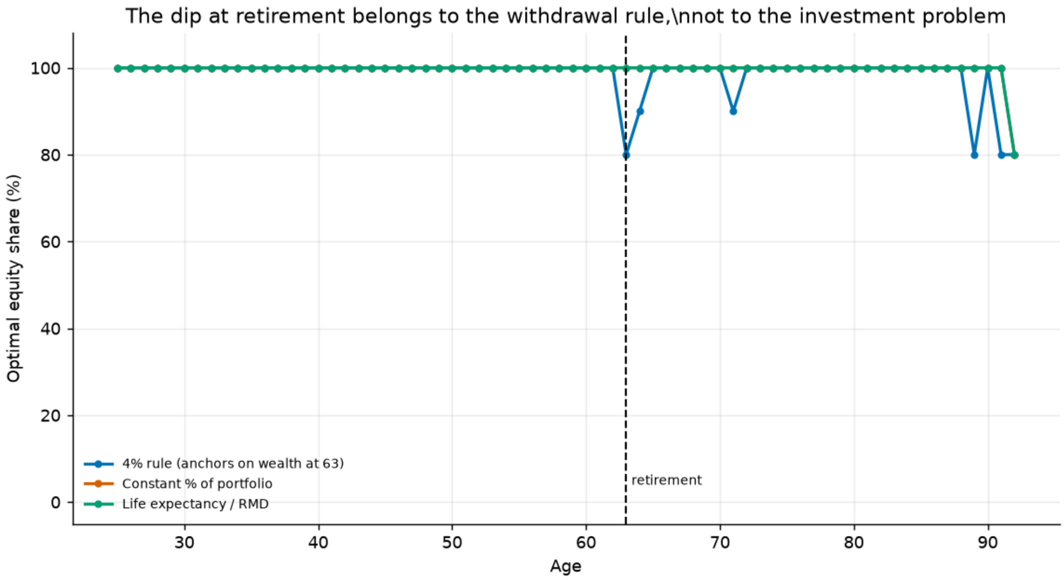

Figure 24.

The cost of anchoring the glide path at retirement. Forcing a conservative allocation at the retirement date is expensive, and the cost rises steeply as the anchor becomes more conservative.

The conclusion of this section is stronger than the one the benchmark comparison supports. It is not merely that the target-date glide path is worse than an all-equity portfolio on this panel; it is that when the schedule is allowed to be anything, the optimiser does not choose a glide path at all.

9. The Whole Allocation, Solved

Section 8 solves for the equity share at every age and for the domestic split on five-year bands, and finds the optimum sits at or near the all-equity corner. But it holds the fixed-income sleeve at a fixed 70/30 bond/bill mix. That restriction was made for search cost, not for principle, and it matters for the interpretation: an optimiser that is told what to hold *inside* a sleeve it barely uses will look more decisive about that sleeve than it actually is.

This section removes the restriction. The decision variable is the full weight simplex at every year of the lifecycle — 68 points in the 3-simplex, 204 free parameters — solved directly against certainty-equivalent consumption. Nothing is held fixed: domestic equity, international equity, bonds and bills compete freely at every age, subject only to non-negativity and to summing to one.

9.1 The search

Under common random numbers the objective is a deterministic function of the schedule, so a search over one age at a time is exact for that age and every sweep is monotone. Two stages are used, because a lattice fine enough to be precise is too large to sweep at every age and a purely local search is too easy to trap:

- a **coarse lattice** sweep, evaluating every composition of the simplex at a step of 25% — the 35 lattice points — at each age in turn; then
- a **fine local** sweep over the twelve single-step pairwise exchanges around the incumbent at a step of 5%, repeated until nothing improves.

A pairwise exchange moves weight from one asset to another and so stays on the simplex by construction, which is what makes it a valid local move there; exchanges that would drive a weight negative are dropped rather than clipped, so every candidate evaluated is feasible. Both stages accept a move only if it clears a relative improvement threshold, for the reason given in Section 4.4.

Table 19. Convergence of the full-simplex search at risk aversion 5

Stage	Sweep	CEC	Gain (%)	Cumulative evaluations
start	-1	0.90340	—	1
coarse	0	1.11231	23.1255	2,381
coarse	1	1.11604	0.3353	4,761
fine	0	1.11888	0.2544	5,111
fine	1	1.12001	0.1013	5,530
fine	2	1.12015	0.0123	5,958
fine	3	1.12030	0.0130	6,383

Notes. The coarse stage does almost all of the work; the fine stage moves the objective by a fraction of a percent. That pattern is itself informative — the surface has a broad flat top.

9.2 The solved schedule

Table 20. The solved four-asset schedule at risk aversion 5, every fifth year

Age	Phase	Dom. equity	Intl. equity	Bonds	Bills	Equity
25	working	0.0%	100.0%	0.0%	0.0%	100.0%
30	working	0.0%	100.0%	0.0%	0.0%	100.0%
35	working	5.0%	95.0%	0.0%	0.0%	100.0%
40	working	0.0%	100.0%	0.0%	0.0%	100.0%
45	working	10.0%	90.0%	0.0%	0.0%	100.0%
50	working	5.0%	95.0%	0.0%	0.0%	100.0%
55	working	10.0%	90.0%	0.0%	0.0%	100.0%
60	working	15.0%	85.0%	0.0%	0.0%	100.0%
65	retired	25.0%	65.0%	0.0%	10.0%	90.0%
70	retired	20.0%	80.0%	0.0%	0.0%	100.0%
75	retired	20.0%	80.0%	0.0%	0.0%	100.0%
80	retired	20.0%	80.0%	0.0%	0.0%	100.0%
85	retired	20.0%	80.0%	0.0%	0.0%	100.0%
90	retired	45.0%	55.0%	0.0%	0.0%	100.0%

Notes. Weights are shares of the whole portfolio and sum to one in each row. The full schedule is in the project's result tables.

Table 21. Average solved weights by lifecycle phase and risk aversion

Risk aversion	Phase	Years	Dom. equity	Intl. equity	Bonds	Bills	Equity
2	working	38	0.0%	100.0%	0.0%	0.0%	100.0%
2	retired	30	12.7%	85.5%	0.0%	1.8%	98.2%
5	working	38	6.1%	93.9%	0.0%	0.0%	100.0%
5	retired	30	13.5%	83.8%	0.7%	2.0%	97.3%
10	working	38	12.9%	85.8%	1.2%	0.1%	98.7%
10	retired	30	13.0%	79.3%	2.8%	4.8%	92.3%

At the baseline preference the solved portfolio averages **9.3% domestic equity and 89.5% international equity**, with 0.3% in bonds and 0.9% in bills. The equity share averages 100.0% through the working life and moves by +0.0 percentage points between its first and last thirds — which is to say it is flat.

The fixed-income sleeve is barely used at any age. That is the direct answer to the question this section exists to ask: the 70/30 bond/bill split Section 8 imposed was fixing the composition of something the optimiser does not want to hold in the first place, so the restriction was not binding. Freeing it buys 0.30% over the best fixed benchmark.

The domestic/international split carries the same caveat as Section 6.2 and should not be read as advice. The international leg in this model is a 37-country leave-one-out average, better diversified than any tradeable international index and available to no individual investor without also holding their own market. A solved schedule that wants 89% international is saying the model prefers *more* diversification than the 50/50 headline strategy provides, not that this particular number is attainable.

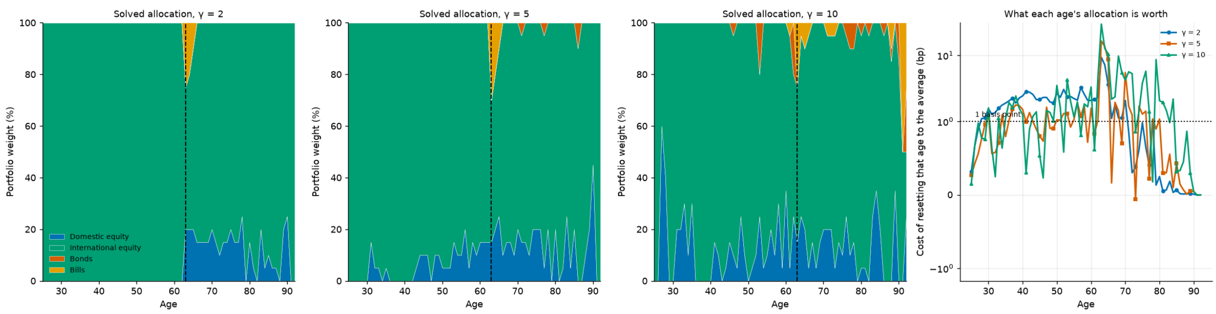

Figure 25.

The solved four-asset schedule at each risk aversion, and what each age of it is worth. The stacked areas are portfolio weights; the right-hand panel resets each age to the schedule's own average and measures the certainty-equivalent cost in basis points, on a log scale with one basis point marked.

9.3 Against the benchmarks

Table 22. Solved and benchmark schedules at risk aversion 5

Schedule	CEC	Gap to best (%)
full simplex optimal	1.1203	0.00
docs07 free form	1.1170	-0.30
docs07 parametric	1.1147	-0.50
100% international equity	1.1060	-1.28
50/50 domestic/international equity	1.0449	-6.73
Target-date fund (glide path)	0.9410	-16.01
100% domestic equity	0.8807	-21.39
60/40 domestic equity/bonds	0.8702	-22.33
100% bills (cash)	0.7380	-34.13

Notes. "Full simplex optimal" is this section's solution. The comparison also includes the schedules solved in Section 8 under the 70/30 restriction, where those are available.

The solved schedule leads the best fixed benchmark (docs07 free form) by **0.30%**. Two things about that number deserve saying plainly.

It is **small**. Freeing 204 parameters and searching them properly buys a fraction of what switching from a target-date glide path to a fixed all-equity portfolio buys in Section 5. The allocation decision has a broad flat top, and almost all of its value is captured by the first choice — whether to hold diversified equity at all — rather than by any refinement of it.

It is also **still not a glide path**. The unrestricted solution differs from the restricted ones of Section 8 in the composition of a sleeve that carries almost no weight, not in the age profile of the equity share. Two independent searches over different parameter spaces reach the same shape, which is a stronger statement than either makes alone.

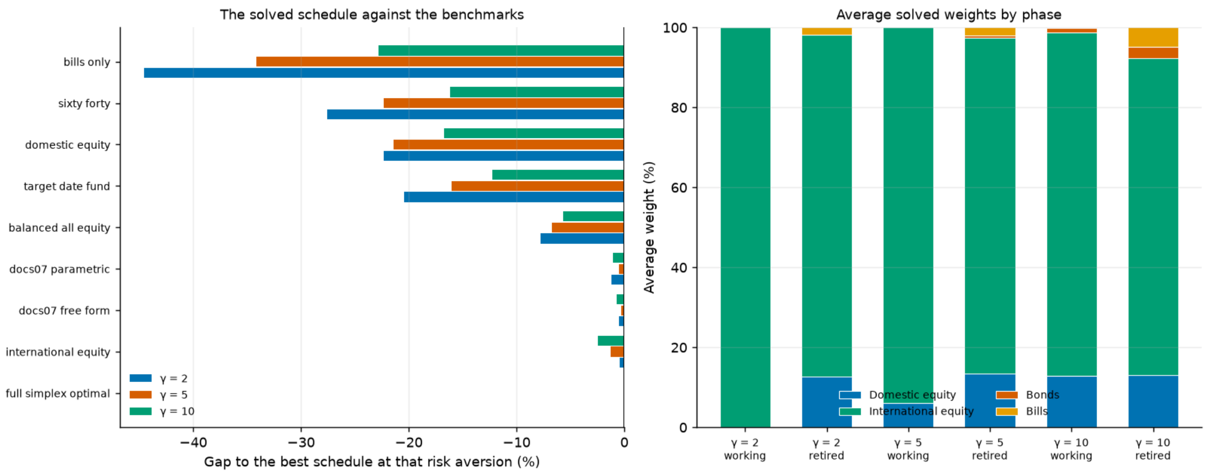

Figure 26.

Left: the solved schedule against the benchmarks at each risk aversion. Right: the average solved weights by lifecycle phase, showing how little of the portfolio the fixed-income sleeve ever carries.

9.4 Where the allocation decision actually matters

A solved schedule always looks structured, and this one looks more structured than the equity-share solve because it has three dimensions to wander in. The deviation profile tests whether the structure is real: each age's allocation is reset to the schedule's own average and the certainty-equivalent cost measured in basis points. Of the 68 ages, **33** move the objective by more than a single basis point.

That is a very different picture from Section 8, where most ages were worth nothing at all, and it is worth reading carefully rather than assuming either conclusion.

Table 23. The eight ages whose allocation is worth the most

Age	Phase	Cost of resetting (bp)	Dom. equity	Intl. equity	Bonds	Bills
63	retired	16.50	15.0%	55.0%	0.0%	30.0%
64	retired	13.66	20.0%	60.0%	0.0%	20.0%
65	retired	8.64	25.0%	65.0%	0.0%	10.0%
70	retired	5.49	20.0%	80.0%	0.0%	0.0%
67	retired	1.94	15.0%	85.0%	0.0%	0.0%
56	working	1.92	20.0%	80.0%	0.0%	0.0%
38	working	1.78	0.0%	100.0%	0.0%	0.0%
58	working	1.77	15.0%	85.0%	0.0%	0.0%

Notes. Cost of resetting that year's allocation to the schedule's own lifecycle average, holding every other year fixed.

The cost is concentrated around the retirement date. The single most valuable age is 63 — the retirement year itself — at 16.5 basis points, and the ten years from 61 to 70 carry 49% of the total cost while being 15% of the lifecycle. That is the sequence-of-returns window Section 12 identifies from a completely different direction, arrived at here through the allocation rather than through the retirement date: what you hold matters most in the years when the portfolio is largest and the withdrawals are about to start.

Working years each cost about 1.1 basis points and retired years 2.0, but the retired average is carried almost entirely by the first few. **The magnitudes remain small in absolute terms** — the largest single age is worth 16.5 basis points and the whole schedule beats the best fixed benchmark by 0.30% — so the honest summary is that the *timing* of the allocation decision is concentrated even though the decision itself is worth little. An investor who got the allocation right in the decade around retirement and used the lifecycle average everywhere else would give up a few basis points.

9.5 Is this a local optimum?

Coordinate ascent cannot escape a local optimum in principle. Re-solving from three different corners of the simplex — an equal split, all equity, and all fixed income — tests for one in practice.

Table 24. Restarting the search from three corners of the simplex

Starting allocation	Solved CEC	Gap to best (%)	Mean dom. equity	Mean intl. equity	Mean bonds	Mean bills
0.25 / 0.25 / 0.25 / 0.25	1.12030	0.0000	9.3%	89.5%	0.3%	0.9%
0.50 / 0.50 / 0.00 / 0.00	1.12012	-0.0156	10.7%	88.5%	0.0%	0.9%
0.00 / 0.00 / 0.50 / 0.50	1.12007	-0.0205	10.1%	88.9%	0.2%	0.8%

Notes. Each restart runs the full two-stage search from the stated starting allocation.

The restarts agree to within **0.021%** of each other. That is not a proof of global optimality — no coordinate search offers one — but it is the check that would have caught the obvious failure, and it did not fire.

10. Borrowing to Invest

Every allocation in this paper so far is long-only and fully invested: the weights are non-negative and sum to one. That is a constraint, not a result, and it rules out a policy with a serious literature behind it. If the case for equities rests on a horizon long enough for diversification across countries and decades to work, then an investor whose financial balance is still small is under-exposed to the very risk they are being told to take, and the natural remedy is to borrow.

The question is never whether leverage raises expected wealth; it obviously does when the expected asset return exceeds the borrowing rate. The question is what it is worth to a *risk-averse* investor at the price they can actually borrow at. This section sweeps that price.

10.1 Mechanics, stated rather than buried

An allocation x over the four assets sums to one. A leverage ratio L means holding L units of that sleeve per unit of equity capital, funding the difference at the real bill rate plus a spread c:

$$r^{p}_{h} = L \cdot (x \cdot r_{h}) - (L - 1) \cdot (r^{bill}_{h} + c) \quad (4)$$

Two choices in that line are load-bearing.

The borrowing rate floats. It is the realised real bill return plus a constant spread, so an investor who borrows is exposed to the same rate their cash would have earned. A fixed real borrowing rate would be a different — and more favourable — assumption.

Limited liability. The portfolio return is clipped at −100%: the lender takes what is left and the investor's equity goes to zero, but they never owe more than they have. That is a margin call, and it is *generous to leverage* — a real levered investor faces forced liquidation at a threshold, not at zero. The tables therefore report how often the clip binds rather than leaving it implicit. The portfolio is rebalanced annually to maintain both the allocation and the leverage ratio, exactly as every unlevered strategy here is rebalanced to maintain its weights.

10.2 Optimal leverage by the price of credit

For each borrowing spread we search jointly over the leverage ratio and the allocation, taking the allocation from the same coarse simplex lattice Section 9 uses. The optimum is therefore a leverage ratio *and* the portfolio that goes with it, rather than a leverage ratio bolted onto a portfolio chosen elsewhere.

Table 25. The optimal leverage ratio and allocation at each price of credit

Borrowing spread	Optimal leverag e	CEC	vs unle vered (%)	Equity	Dom. equity	Intl. equity	Bonds	Bills	Wipeout share
0.00%	1.75x	1.1629	5.10	75%	0%	75%	25%	0%	0.0000%
0.50%	1.25x	1.1426	3.27	100%	25%	75%	0%	0%	0.0000%
1.00%	1.25x	1.1285	2.00	100%	25%	75%	0%	0%	0.0000%
1.50%	1.1x	1.1162	0.88	100%	25%	75%	0%	0%	0.0000%
2.00%	1.1x	1.1105	0.37	100%	25%	75%	0%	0%	0.0000%
3.00%	1x	1.1064	0.00	100%	25%	75%	0%	0%	0.0000%
5.00%	1x	1.1064	0.00	100%	25%	75%	0%	0%	0.0000%

Notes. The spread is an annual real cost over the realised real bill rate. "Wipeout share" is the fraction of path-years in which the limited-liability clip binds — the assumption most favourable to leverage, reported rather than assumed away.

At zero cost, leverage is worth having: the optimum is 1.75× and it is worth +5.10% of certainty-equivalent consumption against the unlevered portfolio.

Note *what* the optimiser levers. At a zero spread it holds 75% equity and levers it 1.75×, for an effective equity exposure of 131% — it borrows against a *diversified* portfolio rather than concentrating into an undiversified one. That is the same preference for diversification the unlevered sections keep finding, expressed through a different instrument.

The interesting number is where it stops. **The break-even spread is 3.00%**: above roughly that annual cost over the real bill rate, no leverage ratio on the grid beats staying unlevered, and the optimum is already 1× by a spread of 3.00%.

That threshold is close to what a household can actually obtain — a retail margin account runs somewhere around one to two percent over the benchmark — so the model does not dismiss leverage out of hand. But the formal break-even overstates how far the case extends. **The advantage falls below a tenth of a percent by a spread of 3.00%**, well before it reaches zero. Over most of the plausible range of borrowing costs, leverage on this panel is not so much *harmful* as *pointless*: it takes on the tail risk documented in Section 10.3 in exchange for a gain that rounds to nothing.

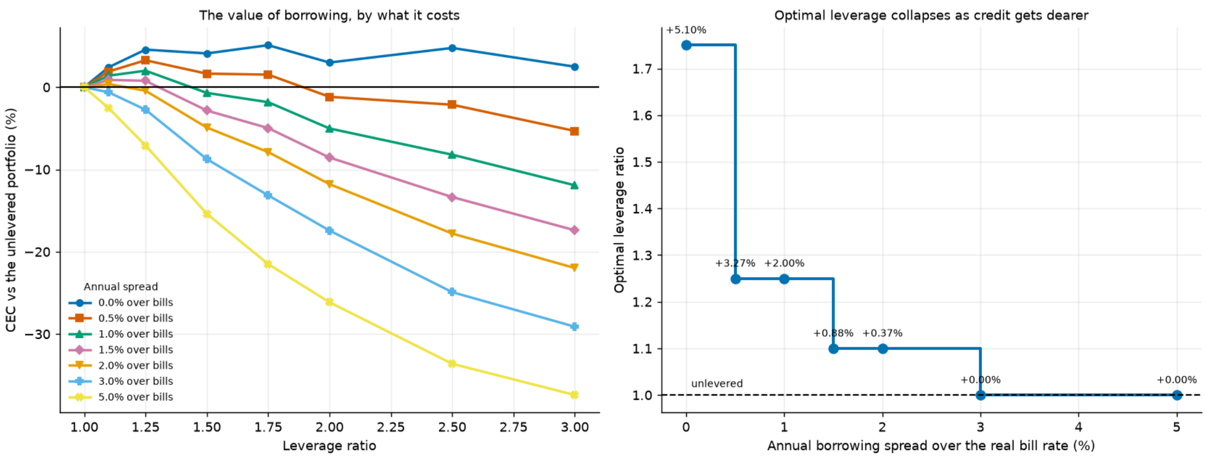

Figure 27.

Left: the value of leverage across the ratio, one line per borrowing spread. Right: the optimal ratio as the price of credit rises, annotated with what it is worth.

10.3 What leverage does to the shape of the outcome

A certainty equivalent alone cannot show what borrowing does to the distribution, and the distribution is the whole argument.

Table 26. Distributional detail at a 1.0% borrowing spread

Leverage	CEC	vs unlevered	P(ruin)	5th pct cons.	Median cons.	95th pct cons.	P(zero bequest)	Wipeout share
1x	1.1064	0.00	8.4%	0.790	1.564	3.701	9.1%	0.0000%
1.5x	1.0987	-0.70	12.6%	0.806	1.951	7.186	13.2%	0.0000%
2x	0.9218	-16.68	24.0%	0.691	2.002	14.004	24.6%	0.0466%
3x	0.7356	-33.52	51.1%	0.539	1.419	37.586	52.0%	1.0090%

Notes. All rows hold the same allocation — the one optimal at that spread — and vary only the leverage ratio, so the differences are attributable to borrowing rather than to reallocation.

Going from unlevered to 3× moves the fifth percentile of retirement consumption by **-31.8%**, the median by -9.3% and the ninety-fifth percentile by +915.5%. The ruin probability moves from 8.4% to 51.1%.

This is the mechanism the certainty equivalent is pricing. Leverage widens both tails, and a risk-averse investor weighs the left one more heavily. The borrowing spread then makes the trade progressively worse, because it is paid in every state of the world including the ones where the leverage did not help.

Note the last column. The clip binds in up to 1.01% of path-years — years in which a real levered investor would have been liquidated rather than merely marked down. It binds only on the high ratios and only when the allocation is held fixed rather than re-optimised, which is why the sweep of Section 10.2 shows none: there the optimiser retreats into bonds and bills precisely to avoid it. Where it does bind, the certainty equivalents overstate what leverage is worth by an amount this model cannot price.

10.4 Should leverage decline with age?

Ayres and Nalebuff (2010) argue that a young investor's financial balance is small relative to the lifetime saving still to come, so a constant *share* of a small balance is a small share of lifetime exposure — and that the remedy is to lever early and delever later. That is a testable claim, and the machinery here tests it directly by solving a leverage ratio at every age, holding the allocation fixed.

Table 27. A leverage ratio solved for every age

Borrowing spread	Leverage at 25	Mean while working	At retirement	Mean in retirement	Solved CEC
0.0%	3x	1.83	1.25x	1.12	1.2263
1.0%	3x	1.73	1x	1.10	1.1766
2.0%	3x	1.57	1x	1.02	1.1428

Notes. The allocation is held at the one optimal for that spread, so the schedule is purely about when to borrow.

The per-age solution is jittery, because the surface is flat enough that the search finds tiny improvements moving a single year between adjacent grid values. Aggregating to decades reports the trend the schedule genuinely carries:

Table 28. Solved leverage by decade of age, at a zero borrowing spread

Decade of age	Years	Mean leverage	Min	Max
20s	5	3.00	3.00	3.00
30s	10	2.10	1.50	3.00
40s	10	1.50	1.50	1.50
50s	10	1.48	1.25	1.50
60s	10	1.12	1.00	1.25
70s	10	1.10	1.00	1.25
80s	10	1.15	1.00	1.25
90s	3	1.17	1.00	1.50

Notes. Aggregated from the per-age solution. The min and max columns show how much the raw schedule moves within each decade — which is where the jitter lives.

The solved schedule **declines with age**, which is the Ayres–Nalebuff prescription arrived at from the other direction: borrow while the financial balance is small relative to the lifetime saving still to come, and delever as it grows. Note that this is the one place in the paper where a genuinely age-varying policy emerges from an unconstrained search — and it is a schedule for *leverage*, not for the equity share. The decade means fall monotonically through the whole working life, so this is not an artefact of the two endpoints.

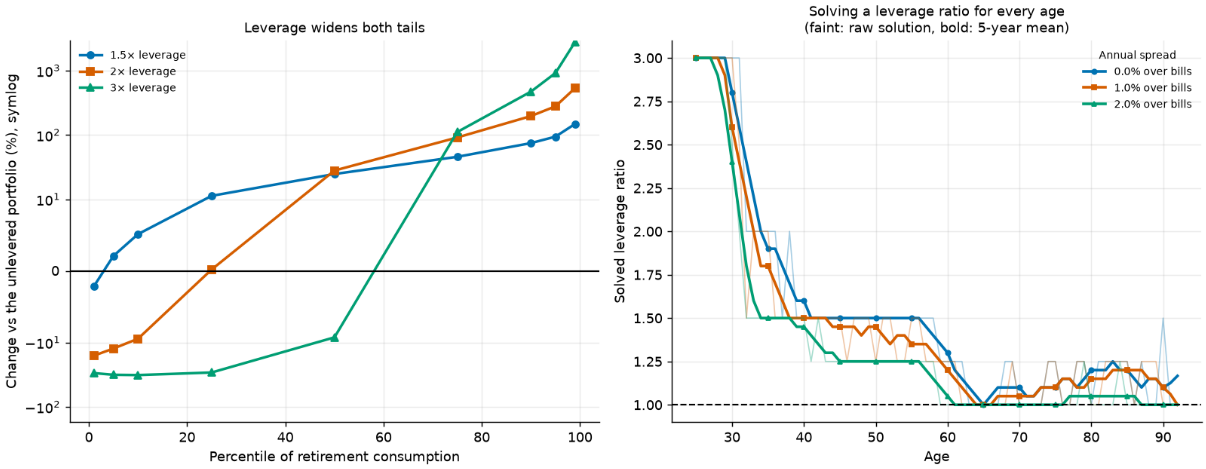

Figure 28.

Left: what leverage does to each percentile of retirement consumption. Right: the leverage ratio solved for every age, at three prices of credit.

10.5 What this changes

- Leverage is **not free money**. It is worth +5.10% when credit is free, breaks even at a spread of 3.00%, and is worth under a tenth of a percent from 3.00% upward — which covers most of the range a household actually borrows in.
- The result is driven by the **left tail**. At a moderate 1.5× the median rises +24.7% while the fifth percentile falls +2.1%; push the ratio to 3× and the median falls too (-9.3%). The certainty equivalent prices that trade at the investor's risk aversion, and a less risk-averse investor would take it at a higher spread.
- What gets levered is a **diversified** portfolio. The optimiser does not respond to cheap credit by concentrating, which is the same answer the unlevered sections give in a different form.
- The limited-liability assumption is **generous to leverage**, so the conclusion is conservative in the direction that matters: a model with forced liquidation would like borrowing less than this one does.

11. Currency Hedging the International Leg

The international leg of the headline strategy is unhedged: a domestic investor holding it bears both foreign equity risk and foreign currency risk. Whether that currency exposure is a cost or a diversifier is an old question, and it has a clean answer in this framework because the panel contains real exchange rates.

We construct a hedged international leg using covered interest parity: the hedged return replaces the realised currency movement with the interest-rate differential implied by the two countries' bill rates, less an explicit annual hedging cost. The cost is the parameter of interest. Hedging is not free — it consumes bid-offer, collateral and roll — and the question a practitioner faces is not "is hedging good?" but "is hedging good at the price I can get it for?" Setting the cost to zero therefore gives the decision its most favourable possible reading, which is where we start.

Table 29. Break-even annual hedging cost by hedge ratio

Hedge ratio	Unhedged CEC	CEC at zero cost	Gain at zero cost (%)	Break-even annual cost
25%	1.049	1.047	-0.135	never
50%	1.049	1.038	-1.021	never
75%	1.049	1.023	-2.432	never
100%	1.049	1.004	-4.240	never

Notes. The break-even cost is the annual charge at which hedging exactly offsets its benefit. "Never" means the hedged position loses even at zero cost, so no price makes it worthwhile.

The answer here is that no price is low enough. Every hedge ratio tested loses certainty-equivalent consumption before a single basis point of cost is charged. The mildest is a 25% hedge, which gives up 0.13%; the fully hedged position gives up 4.24%. The loss grows monotonically in the ratio, so there is no break-even cost to report: the decision is settled at zero, and adding a realistic charge only widens the gap.

Tracing the optimal ratio as a function of cost adds nothing, because it is 0% at every cost including free. For a lifecycle investor holding this international leg, the model's answer is not “hedge if you can get it cheaply” but “do not hedge.”

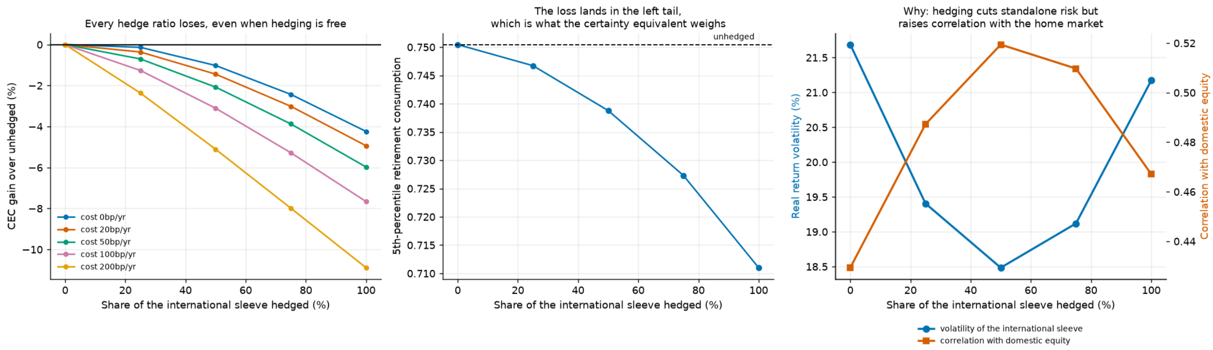

Figure 29.

Left: the certainty-equivalent cost of hedging by ratio, at five annual hedging costs; every line is below zero, so hedging loses even when free. Centre: where the loss lands — fifth-percentile retirement consumption falls monotonically in the hedge ratio, and the certainty equivalent weighs that tail heavily. Right: why — hedging lowers the standalone volatility of the foreign sleeve up to a half hedge, but raises its correlation with the home market over the same range.

12. Endogenous Retirement Timing

Every result so far retires the investor on a birthday. Real people do not. They retire when the balance looks big enough — which means, mechanically, that they retire disproportionately after good markets. That is worth testing rather than assuming, because a balance that looks big after a bull run is also a balance bought at high valuations, and the same market move that triggers the decision may be lowering the returns that have to fund it.

This section makes the retirement date a path-dependent decision and asks two questions: whether a wealth trigger beats a fixed date, and how much of the outcome the decade around the retirement date explains.

12.1 A wealth trigger against a date

Utility here is evaluated over the whole lifetime rather than the retirement window, because a rule that retires people early buys them leisure that a retirement-only window would charge for without crediting. We also introduce a working-life income floor of 25% of average earnings. That is not decoration either: retirement in this model carries a real consumption floor through the progressive social-security schedule and working life, at zero, carries none. Wherever compared policies share the same working-life consumption the asymmetry cancels, but here it does not — and left uncorrected it rewards retiring early purely for reaching the safety net sooner.

Table 30. Fixed retirement dates against wealth triggers

Rule	CEC γ=5	Median age	Mean age	S.d. age	P(ruin)	Median wealth at retirement	Median cons.
Fixed age 60	0.804	60.000	60.000	0.000	14.0%	17.3	1.325
Fixed age 63 (baseline)	0.790	63.000	63.000	0.000	11.7%	21.3	1.485
Fixed age 66	0.772	66.000	66.000	0.000	9.2%	25.9	1.674
Fixed age 70	0.745	70.000	70.000	0.000	5.8%	33.6	1.975
Flexible +/-3 years, 25x income	0.799	65.000	63.755	2.505	11.5%	23.3	1.552
Wealth trigger 20x income	0.815	62.000	62.743	5.368	12.4%	22.0	1.493
Wealth trigger 25x income	0.807	65.000	64.223	5.213	11.1%	24.3	1.582
Wealth trigger 30x income	0.799	67.000	65.336	4.929	10.3%	26.2	1.659

Notes. A wealth trigger retires the investor once financial wealth reaches the stated multiple of current income, within an age window. Whole-lifetime utility, working-income floor at 25% of average earnings.

Read naively this table says wealth triggers beat fixed dates. That reading is wrong, and the reason is instructive enough to spend a paragraph on. The model contains no disutility of labour: working an extra year is costless and produces income, so the certainty equivalent falls monotonically with the fixed retirement age for purely mechanical reasons. A wealth trigger that retires people later on average will therefore look better than age 63 without any conditioning value at all.

The correct comparison holds the mean retirement age fixed. We interpolate the fixed-date frontier at each rule's own realised mean retirement age and score against that.

Table 31. The value of conditioning, against a date matched on the same mean age

Rule	Mean retirement age	CEC	Matched fixed-date CEC	Value of conditioning (%)
Flexible +/-3 years, 25x income	63.755	0.799	0.786	1.65
Wealth trigger 20x income	62.743	0.815	0.791	3.03
Wealth trigger 25x income	64.223	0.807	0.783	3.11
Wealth trigger 30x income	65.336	0.799	0.776	2.98

Notes. This is the number that isolates conditioning from timing. The unmatched comparison in the previous table conflates the two.

Matched, the value of conditioning is 3.11% at the best trigger — real, but roughly 62% smaller than the same comparison without the working-income floor, where it reaches 8.14%. Both numbers are reported because the difference between them is a model artefact we found and removed, not a result.

12.2 The retirement-date lottery

How much of a retirement outcome is decided by when you happened to be born? We regress the outcome on the annualised real portfolio return over the decade centred on each path's own retirement date. That window explains 5.9% of the variation in retirement consumption, against 16.8% for the whole 68-year lifetime. A single decade — 15% of the horizon — carries 35% of the explanatory power of the whole thing.

That is the sequence-of-returns problem stated as a number, and it is larger than the difference between any two allocation strategies in this paper. It is also the strongest argument for the flexible retirement date of the previous subsection: the one lever that acts directly on the lottery is the ability to choose which decade you retire into.

Table 32. Retirement outcomes by decile of the retirement-decade return

Decile	Mean Window Return	Median Retirement Consumption	Prob Ruin	Median Retire Age	Median Wealth At Retirement
1.000	-0.028	1.229	0.345	70.000	17.309
2.000	0.017	1.446	0.191	67.000	21.275
3.000	0.039	1.477	0.137	66.000	21.840
4.000	0.056	1.546	0.109	65.000	23.221
5.000	0.070	1.629	0.085	65.000	25.040
6.000	0.084	1.682	0.083	64.000	26.286
7.000	0.098	1.680	0.064	64.000	26.239
8.000	0.114	1.708	0.049	64.000	26.810
9.000	0.132	1.709	0.030	64.000	26.865
10.000	0.172	1.814	0.019	63.000	29.336

Notes. Paths sorted by the annualised real return over the ten years centred on their own retirement date, then split into deciles.

Do people retire into bull markets, and does it hurt?

Under a wealth trigger the answer to the first question is yes: the correlation between retirement age and the preceding market run-up is -0.310, so a strong market pulls the retirement date forward. Early retirees experience a mean pre-retirement run-up of 12.1% against 7.1% for late ones.

The second question is the interesting one, and the answer is reassuring for the trigger. The correlation between the run-up and the *subsequent* return is -0.068 — essentially zero. On this panel, retiring after a good decade does not systematically buy you a bad one. The folk warning that bull markets lure people into retiring at the worst possible time is not supported here: the mean subsequent return for early retirees (6.0%) is if anything slightly higher than for late ones (6.0%). We report this because it cuts against the intuition that motivated the test.

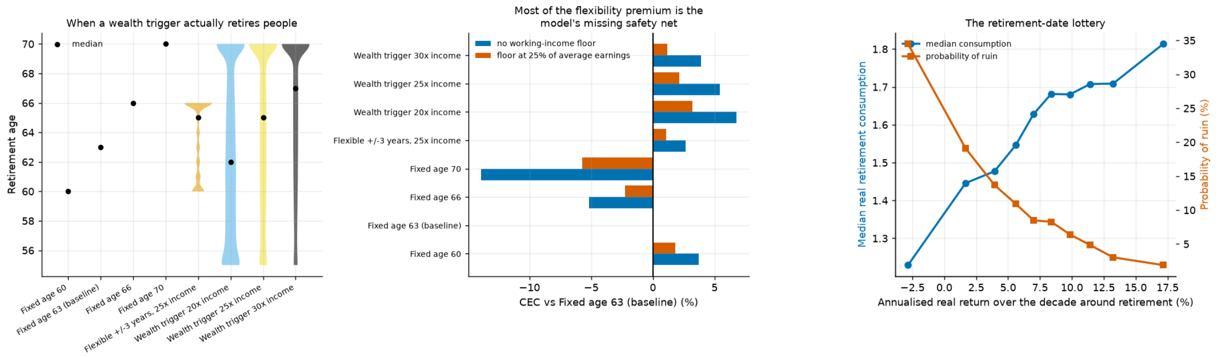

Figure 30.

Endogenous retirement timing. The distribution of retirement ages under a wealth trigger, the value of conditioning against a matched fixed date, and the retirement-date lottery.

13. Conditioning the Savings Rate

Section 12 found that conditioning *when* you stop working on your financial position is worth something. This section asks the accumulation-side mirror: should the savings rate vary over a career, and should it respond to the portfolio?

Two quite different things could make the rate vary, and a comparison that does not separate them will credit the wrong one. **Shape** is variation with age alone: labour income here is hump-shaped, and a fixed rate makes consumption track income exactly, so a 25-year-old consumes least precisely when they are poorest. **Conditioning** is variation with state: whether wealth is ahead of or behind an age-appropriate target. The first uses no market information at all; the second is what "you should have six times salary by fifty" is reaching for.

13.1 A caveat that has to come first

The model cannot identify the savings *level*. Swept continuously, the certainty equivalent peaks at a constant rate of 5% and falls above it. That number should not be believed and it is worth being explicit about why.

The savings level is a trade between consuming now and consuming later. In this model that trade is settled almost entirely by the discount factor — $\beta = 0.96$, which over a 38-year working life discounts retirement consumption by a factor of 0.21 — and by risk aversion acting on the left tail of consumption, which with a floored retirement and risky labour income sits in working life rather than in retirement. Neither of those is something a panel of historical returns has any view on.

So the level is not identified here and this paper does not claim it. What the return panel *can* speak to is the shape of the profile and the value of conditioning it, both holding the average rate fixed. Everything below pins the career-average savings rate at 10% and asks only when, and on what, to save it.

13.2 Shape: when should you save?

We solve for a free savings rate at each of the 38 working years by coordinate ascent over a per-age multiplier, renormalising after every move so that the career average never drifts. The answer is not the same at every risk aversion, and that is the interesting part.

Table 33. Solved savings profile by risk aversion, career average pinned

Risk aversion	Shape	First quarter	Middle half	Last quarter	Peak age	Gain vs flat (%)
γ = 2	hump-shaped	3.5%	13.2%	9.3%	38	0.48
γ = 5	hump-shaped	5.8%	13.8%	6.1%	35	0.62
γ = 10	front-loaded	15.8%	11.7%	1.3%	25	2.48

Notes. The profile is summarised by the average rate in the first quarter, middle half and last quarter of the working life. All three profiles have the same career average by construction, so the comparison is purely about shape.

At moderate risk aversion the solved profile is **hump-shaped**: save least when young, most in peak-earning years, taper into retirement. That is consumption smoothing doing what theory says it should. A 25-year-old sits at the bottom of a hump-shaped income profile, and taking a further tenth of that income away is expensive in utility terms precisely because there is so little of it. This is the "save more later, when you earn more" pattern, and the model produces it without being told to.

At high risk aversion it **inverts** to front-loaded. Two motives compete and risk aversion picks the winner. Smoothing wants saving where income is highest, which is mid-career. Precaution wants a buffer built early, so that it has the whole remaining career to compound against a bad income or market draw. At γ = 10 precaution wins outright. The model cannot settle which kind of investor a given reader is; what it can say is that both beat a flat rate and that getting the shape right matters more the more risk-averse you are.

Unlike the glide path of Section 8, this structure is real. The deviation profile shows 32 of 38 working years moving the objective by more than a basis point when reset to the career average.

13.3 Conditioning: your position beats the market's direction

Layered on top of the solved shape, and scored against a constant rate interpolated at each rule's own realised career average, we test two conditioning rules: an *on-track* rule that raises the rate when wealth is below an age-appropriate wealth-to-income target, and a *return-responsive* rule that raises it after a weak market year.

Table 34. Conditioning rules at a matched career-average savings rate

Rule	Realised mean rate	CEC	Matched constant CEC	Value (%)
On-track k=0.05	10.62%	0.817	0.783	4.32
On-track k=0.02	10.02%	0.813	0.787	3.29
On-track k=0.01	9.83%	0.808	0.788	2.51
On-track k=0.005	9.81%	0.803	0.789	1.88
Solved shape	10.00%	0.792	0.788	0.62
On-track k=0	10.00%	0.792	0.788	0.62
Return-responsive k=0	10.00%	0.792	0.788	0.62
Return-responsive k=0.25	8.46%	0.799	0.796	0.45
Return-responsive k=-0.25	11.93%	0.773	0.773	-0.00
Return-responsive k=0.5	7.87%	0.794	0.799	-0.59

Notes. The matched comparison is essential: a rule that saves more when behind will, in a world where most paths fall behind, quietly save more overall. Interpolating the constant-rate frontier at each rule's own realised mean strips that out.

Saving more when behind an age-appropriate wealth target is worth several times what the age profile alone is worth. Saving more after a bad market year is worth almost nothing. The sign check behaves as it must: reversing the on-track rule — saving *less* when behind — is strongly negative, which is the reassurance that the machinery measures what it claims.

The interpretation is that a bad market year is a poor proxy for being behind. Wealth relative to an age-appropriate target is the sufficient statistic; the market's recent direction is a noisy shadow of it. Section 14 takes this apart in detail and finds that even this reading needs qualifying.

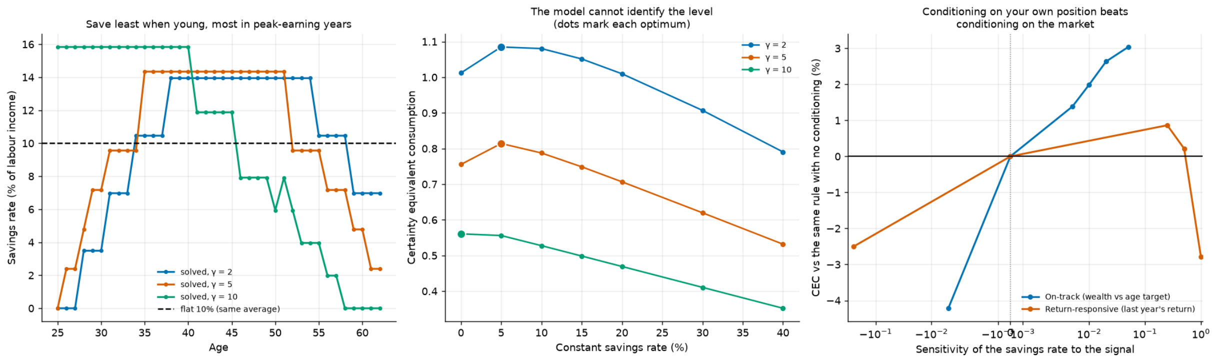

Figure 31.

The solved savings profile by risk aversion, the unidentified constant-rate frontier, and the value of conditioning on wealth against conditioning on returns.

13.4 Do the savings and retirement gains add?

Section 12 conditions the retirement date on the portfolio; this section conditions the savings rate on it. A natural question is whether an investor should do both.

Table 35. Savings conditioning and retirement conditioning, separately and together

Configuration	CEC	vs neither (%)	Mean rate	Mean retirement age	P(ruin)
Neither	0.788	0.00	10.00%	63.000	11.7%
Savings conditioning only	0.817	3.68	10.62%	63.000	11.7%
Retirement conditioning only	0.805	2.20	10.00%	64.203	11.1%
Both	0.814	3.34	11.71%	63.346	11.8%

Notes. Whole-lifetime utility. Both rules read the same underlying signal, so their gains overlap rather than add.

They do not add. Savings conditioning alone is worth 3.68%, retirement conditioning alone 2.20%, and both together 3.34% — less than the better of the two alone. Both rules respond to the same ahead-or-behind signal, so running them together over-corrects. This is a practical result: an investor choosing between the two levers should choose, not stack.

14. The Accumulation Signal, Decomposed

Section 13 established that conditioning the savings rate on the funded ratio is worth roughly three percent of certainty-equivalent consumption. That result rests on five choices that were made once and never varied: the gap was measured in income multiples, the target was the model's own median path, one coefficient governed both directions, the rate was free to move anywhere between 0% and 40%, and the rule ran for the whole career. Each is a modelling decision, and a number that survives only one setting of them is not worth acting on.

This section varies all five, races five further state variables against the funded ratio, tests whether the two best combine, and asks where in the distribution and at which ages the value lands. 195 rule evaluations at 20,000 paths each, all under common random numbers.

Two scoring conventions apply throughout and both matter. Each number is a percentage gain over a *constant* savings rate interpolated at the rule's own realised career average. And conditioning is layered on top of a solved age profile that is itself worth +0.62% against that matched constant, so every figure quoted below is *net* of the shape's own contribution.

14.1 Which units is "behind" measured in?

Being "two times salary short" means something entirely different at 30 and at 60, because the target itself grows from roughly nothing to around ten times income over a career. A response linear in that gap is therefore inert when young and violent when old, whether or not anyone intended it. We test two scale-free alternatives that measure the shortfall as a fraction of the target instead.

Table 36. Three ways of measuring the shortfall, each at its own best coefficient

Gap measured as	Best k	Extra saving when 25% behind (pp)	Value (%)	Net of shape (%)	Mean rate	P(ruin)
Funded ratio (1 − W/Y ÷ target)	0.400	10.00	5.05	4.43	9.56%	11.7%
Log funded ratio (log target ÷ W/Y)	0.200	5.75	5.02	4.40	9.63%	11.7%
Level gap (target − W/Y, income multiples)	0.100	8.16	4.40	3.78	11.22%	11.7%

Notes. The three forms measure the gap in different units, so their coefficients are not comparable. The third column translates each into a common scale: percentage points of income added for a path a quarter short of target at mid-career.

Ranked at their own optima the two scale-free forms are indistinguishable and both beat the level gap. But that comparison is not quite fair, because the three grids do not reach equally far. Interpolating every curve at the strongest response *all* of them can produce separates "this form is better" from "this form's grid let it go further".

Table 37. Functional forms compared at matched response strength

Form	At the same move (pp)	Value there (%)	Best anywhere on its grid (%)	Furthest its grid reaches (pp)
proportional	10.00	5.05	5.05	10.00
log	10.00	4.86	5.02	11.51
level	10.00	4.32	4.40	12.23

Notes. Each form's curve interpolated at the largest contribution move that all three grids can produce. This is the comparison the text relies on; the previous table's ranking partly reflects how far each grid reaches.

At matched strength 0.73 percentage points still separate the forms, so the ranking is about the shape of the response and not only about its strength. A rule linear in income multiples is inert exactly when the investor has time to respond and violent exactly when they do not, and it pays for that.

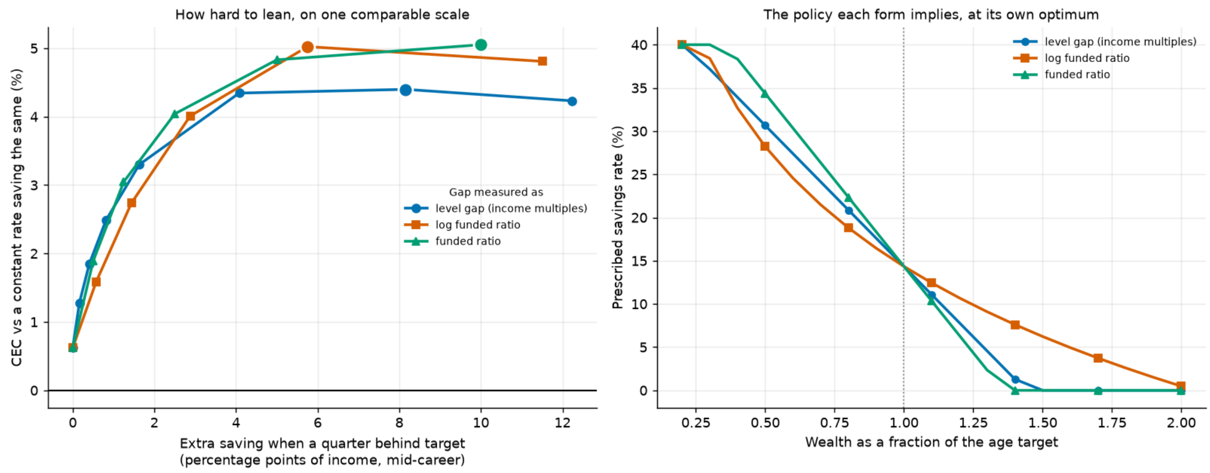

Figure 32.

Left: the value of conditioning against the contribution move it implies, which puts three differently-scaled coefficients on one axis. Right: the policy each form implies at its own optimum, over the range of funded ratios paths actually visit.

14.2 Catching up and easing off are separate policies

A symmetric rule does two things at once: it saves more when behind and less when ahead. There is no reason those should carry the same coefficient, and separating them is the question a real saver faces — almost all published advice is about catching up, and almost none is about easing off.

Table 38. Which half of the conditioning rule earns its keep

Rule	k behind	k ahead	Value (%)	Net of shape (%)	Realised mean rate
Neither (age profile only)	0	0	0.62	0.00	10.0%
Catch up only (never ease off)	0.40	0	3.09	2.47	12.8%
Ease off only (never catch up)	0	0.40	3.25	2.63	7.4%
Both, free coefficients	0.40	0.10	5.26	4.64	10.5%

Notes. Each half is optimised separately over the same grid. Note the realised mean rates: catch-up alone raises the career average well above target and ease-off alone cuts it well below, so only the matched-rate comparison makes them comparable.

Net of the age profile, catching up alone is worth +2.47% and easing off alone +2.63% — a ratio of 1.06, which on any reasonable reading is a tie. That is itself the finding, because the two halves are not symmetric in anything except their value: catch-up alone raises the career savings rate to 12.8% and ease-off alone cuts it to 7.4%. They arrive at the same place from opposite directions. On raw certainty equivalent the one that saves more would simply look better; only the matched comparison reveals the tie.

Run together they reach +4.64%, against 5.10% for the two summed. The halves are strongly sub-additive: much of what each earns alone is the same correction, arrived at from opposite sides. The free search picks different coefficients, so there is no case here for a deliberately asymmetric rule.

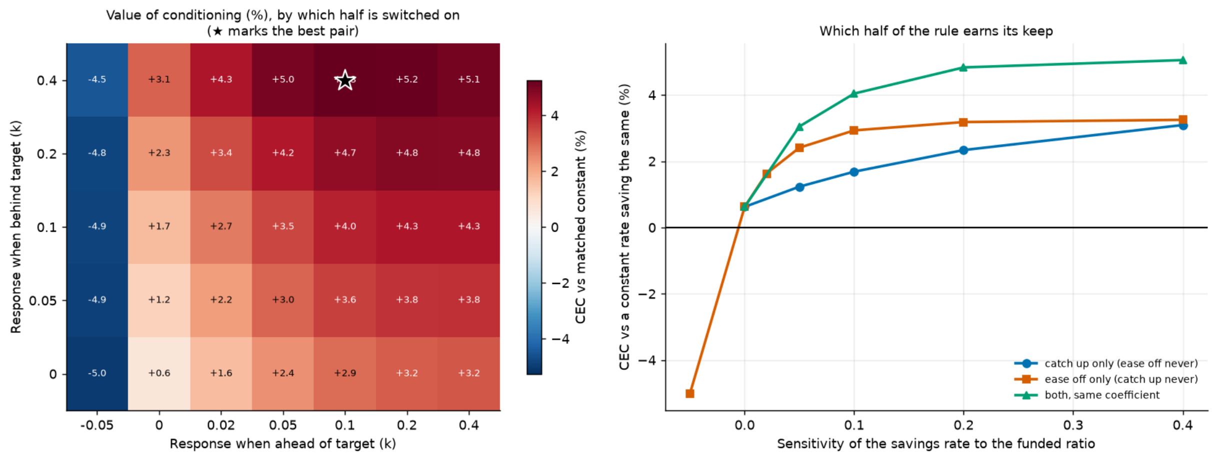

Figure 33.

Left: the value of conditioning over the two coefficients separately, with the best pair starred. Right: each half of the rule switched on alone against both together.

14.3 The version a person could actually follow

A continuous response asks for a freshly computed contribution rate every year. Section 7 found that coarse guardrail rules give up surprisingly little on the spending side; the accumulation-side analogue is to check once a year and move the contribution by a fixed step only if the funded ratio is more than a dead band away from target.

Table 39. Guardrail savings rules

Dead band (± of target)	Rate step	Value (%)	Net of shape (%)	Mean rate	P(ruin)
10%	5%	3.20	2.58	9.70%	11.7%
20%	5%	3.14	2.52	9.67%	11.7%
35%	5%	2.85	2.23	9.60%	11.7%
10%	3%	2.36	1.73	9.85%	11.7%
20%	3%	2.29	1.67	9.82%	11.7%
35%	3%	2.09	1.47	9.76%	11.7%
10%	2%	1.85	1.23	9.91%	11.7%
20%	2%	1.79	1.17	9.89%	11.7%

Notes. A two-number rule: check the funded ratio once a year, and move the contribution by the step only if the ratio is more than the dead band away from target.

The best guardrail — move 5% of income once you are more than 10% off target — is worth +2.58%, or 58% of what the tuned continuous rule earns. It wants the narrowest dead band and the largest step the grid offers, which is the search saying it would rather be continuous; the table prices that compromise rather than pretending there is none. Even so, keeping most of the value for a rule with two numbers and no arithmetic is the trade most savers should take.

14.4 Does the target have to be right?

The target is the weakest link in the construction: it is the model's own median wealth path, which no investor could know in advance. We test two alternatives — a published "N times salary by age X" ladder of the kind large fund managers publish, and a flat multiple with no age content at all.

Table 40. Three wealth-to-income targets, unscaled

Target	Median multiple asked for	Value (%)	Net of shape (%)	Realised mean rate
model median path	11.4	5.05	4.43	9.6%
published ladder	7.7	1.76	1.14	9.6%
flat 8x income	8.0	-7.47	-8.09	15.6%

Notes. The ladder is a published rule of thumb; the flat multiple has no age content and serves as a deliberate straw man.

The published ladder captures 26% of the model-implied target's value — useful, but not a substitute. The flat multiple is worth -8.09%: **worse than not conditioning at all**. Telling a 28-year-old they should already hold eight times salary leaves them behind target for most of a career and drives the career average savings rate far above the pinned target. That is not conditioning, it is just saving more, and the matched comparison charges for it.

The ranking says two separate things. A target that rises with age is necessary — the flat one is not merely weaker, it is harmful. But rising is not sufficient: the ladder rises and still gives up a substantial share of the value, so how it rises matters too.

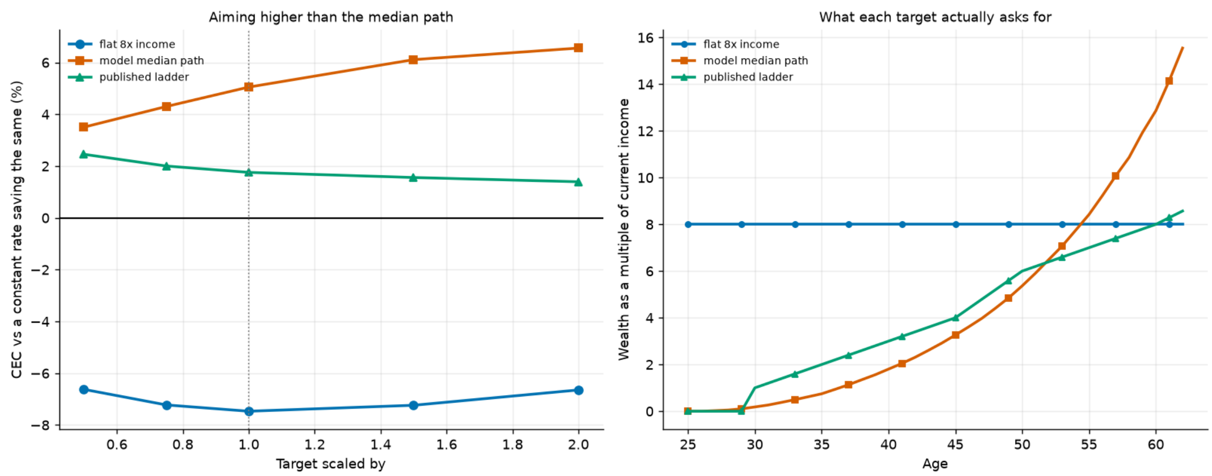

Figure 34.

Left: the value of conditioning as each target is scaled up and down. Right: what each target actually asks the investor to hold at each age.

14.5 Which signal, out of everything available?

Eight candidate state variables, all signed so that positive means "save more", all bounded so that one sensitivity grid is meaningful across them, all swept over the same grid and scored the same way. They fall into three kinds: *stock* signals that say where the investor is, *flow* signals that say what the market just did, and the pay cheque.

Table 41. The signal horse race

Signal	Kind	Best k	Value (%)	Net of shape (%)	Mean rate	P(ruin)	5th pct cons.
Income vs its expected path	income	0.40	7.07	6.45	11.3%	11.9%	0.522
Funded ratio (wealth vs age target)	stock (balance vs target)	0.40	5.05	4.43	9.6%	11.7%	0.772
Raw balance (wealth ÷ income)	stock (balance)	0.40	2.08	1.46	4.2%	11.7%	0.626
Investment gain (balance vs contributions)	stock (market's share of the balance)	0.40	0.97	0.35	1.3%	11.7%	0.489
Trailing 5-year return	flow (market)	0.10	0.65	0.03	9.4%	11.7%	0.722
Trailing 10-year return	flow (market)	0.10	0.65	0.03	9.4%	11.7%	0.720
Last year's return	flow (market)	0.02	0.63	0.01	9.9%	11.7%	0.730
No conditioning (age profile only)	none	0.00	0.62	0.00	10.0%	11.7%	0.732

Notes. Each signal at its own best sensitivity. The "no conditioning" row appears once for every point on the grid and spans zero percentage points across them, which is the determinism check that the common random numbers are holding.

The winner is not a portfolio signal at all. Saving more in years when the pay cheque runs above its expected path — and less when it runs below — is worth +6.45%, against +4.43% for the funded ratio and +1.46% for the raw balance. This is the accumulation-side version of consumption smoothing: an income shock is observed *before* it is spent, so responding to it costs the investor very little utility, whereas responding to a portfolio shortfall means cutting consumption that was already planned. It is also the easiest signal in the list to act on, because it needs no target, no balance and no arithmetic.

Every flow signal is worth essentially nothing. Trailing one-, five- and ten-year returns come in at +0.01%, +0.03% and +0.03% respectively. At that size the ordering within the group is not meaningful and should not be read as one. A return signal knows what the market did but not whether it left *this* investor short — it is the same market for a 30-year-old with nothing saved and a 60-year-old with ten times salary, and they should not respond alike.

Layering the two leaders together reaches +6.92%, against +6.45% for the better one alone and 10.88% if they were additive. They overlap — a run of weak income and a run of weak markets both show up as a balance behind target — but they do not overlap completely, because only one of them is visible before the money is spent.

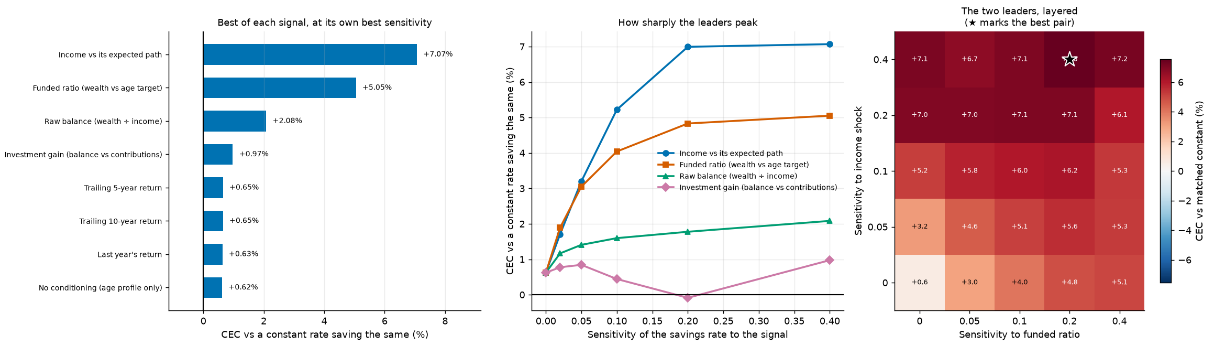

Figure 35.

Left: the best of each signal at its own optimal sensitivity. Centre: how sharply the leaders peak. Right: the two leaders layered, showing partial but incomplete overlap.

14.6 How far does the contribution have to move?

The unconstrained optimum may send the savings rate anywhere in the permitted range. No household budget works like that. Re-pricing the rule with the rate confined to progressively narrower bands around its average is the cheapest test of whether the finding survives contact with a real household.

Table 42. The value of conditioning under a constrained contribution

± allowed move	Floor	Cap	Net value (%)	Share of un constrained (%)	Mean rate	P(ruin)
0%	10%	10%	-0.62	-14	10.0%	11.7%
1%	9%	11%	-0.05	-1	9.7%	11.7%
2%	8%	12%	0.54	12	9.5%	11.7%
3%	7%	13%	1.14	26	9.5%	11.7%
5%	5%	15%	2.24	51	9.4%	11.7%
8%	2%	18%	3.36	76	9.3%	11.7%
10%	0%	20%	4.06	92	9.1%	11.7%
15%	0%	25%	4.31	97	9.4%	11.7%
30%	0%	40%	4.43	100	9.6%	11.7%

Notes. The rate is confined to the career average plus or minus the stated width. At zero width the rule collapses onto the constant rate it is scored against, which is why the first row is exactly zero.

There is no cheap corner here. Up to about ten points the value is close to proportional to the flexibility given — ±3 points of income buys 26% of the unconstrained value, ±5 points 51%, ±10 points 92% — and only then does it saturate. A household that can flex its contribution by a couple of points does *not* get most of the benefit; it gets roughly its share. That is the least convenient version of this result and it is the one the table supports.

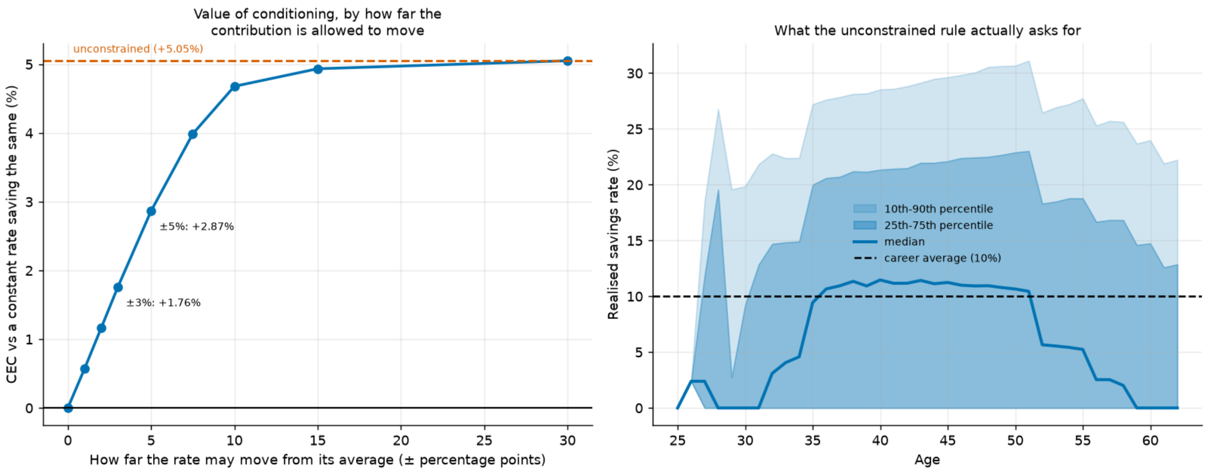

Figure 36.

Left: the value of conditioning as a function of how far the contribution may move. Right: the distribution of prescribed savings rates by age under the unconstrained rule.

14.7 Where the gain lands, and who wants it

A certainty equivalent is one number. It cannot say whether a rule lifted the middle of the distribution or insured the bottom of it, and those are very different products.

Table 43. Retirement consumption by quantile, with and without conditioning

Quantile	No conditioning	Conditioned	Change (%)
p1	0.579	0.604	4.43
p5	0.732	0.772	5.40
p10	0.835	0.892	6.86
p25	1.077	1.153	7.07
p50	1.459	1.582	8.43
p75	2.018	2.224	10.20
p90	2.831	3.085	8.98
p95	3.514	3.821	8.75
p99	5.417	5.703	5.28

Notes. Average real retirement consumption per path, sorted. The rule is funded by contributions taken from consumption on the paths that were going to be comfortable, which is why the top percentile is the one place the change turns negative.

This is not left-tail insurance, and it is not a free lunch either. The gain is positive across almost the whole distribution and is largest in the middle (+8.57% against +5.56% at the bottom decile and +7.67% at the top). The bottom of the distribution is where the rule has least to work with — a path that is behind *because* labour income collapsed cannot save its way out — and the top is where the rule takes its payment. We report this because the intuition that motivated the test was that conditioning would be tail insurance, and it is not.

Across preferences the value rises steeply with risk aversion: +2.13% at γ = 2, +4.43% at γ = 5 and +5.58% at γ = 10. Every preference picks the same coefficient, so what changes is not the rule but how much the investor will pay for it. Conditioning is a risk product, and its price is set by how much the buyer dislikes risk.

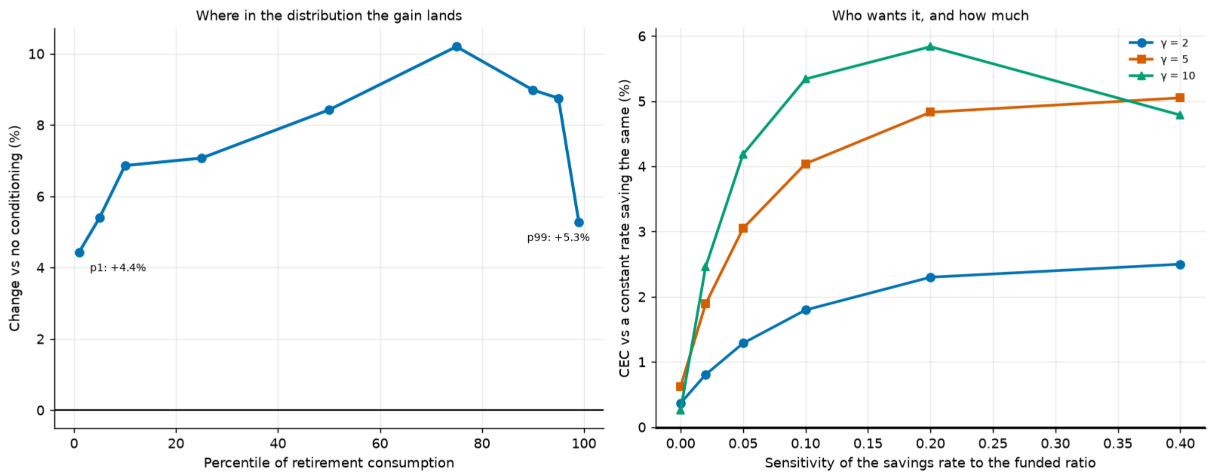

Figure 37.

Left: where in the distribution of retirement consumption the gain lands. Right: the value of conditioning by risk aversion, each at its own coefficient.

14.8 When in a career is the balance worth reading?

Switching the conditioning on only for part of the career prices each stretch of working life separately. Outside the window the rule falls back to the age profile exactly, so these are clean subtractions. The sweep includes overlapping spans because halves and thirds answer different questions; the text below uses only the three non-overlapping windows that tile the career, because comparing a 25-year window against a 13-year one says more about length than about timing.

Table 44. The value of conditioning by career stretch

Conditioning active for ages	Net value (%)	Mean rate	P(ruin)
25-62	4.43	9.6%	11.7%
40-62	3.35	10.2%	11.7%
25-49	2.69	9.1%	11.7%
50-62	2.13	10.9%	11.7%
40-49	1.58	9.7%	11.7%
25-39	1.19	9.6%	11.7%

Notes. Each row switches the rule on only for the stated ages. The three non-overlapping windows tile the career and are the ones the discussion relies on.

The value rises monotonically with age across the three tiles. The balance is close to uninformative when there is barely any of it and decades left to recover; it becomes informative once a shortfall is large in absolute terms and there is little time left to fix it. The three tiles sum to approximately what the whole career is worth, so unlike the two directions of Section 14.2 these stretches really are close to separable.

How hard the rule leans at each age is a different question from whether leaning there is worth anything, and the two do not track each other one for one: per year of conditioning the value varies more across the career than the size of the adjustment does. The early career is not just quieter, it is worth less than its quietness would suggest.

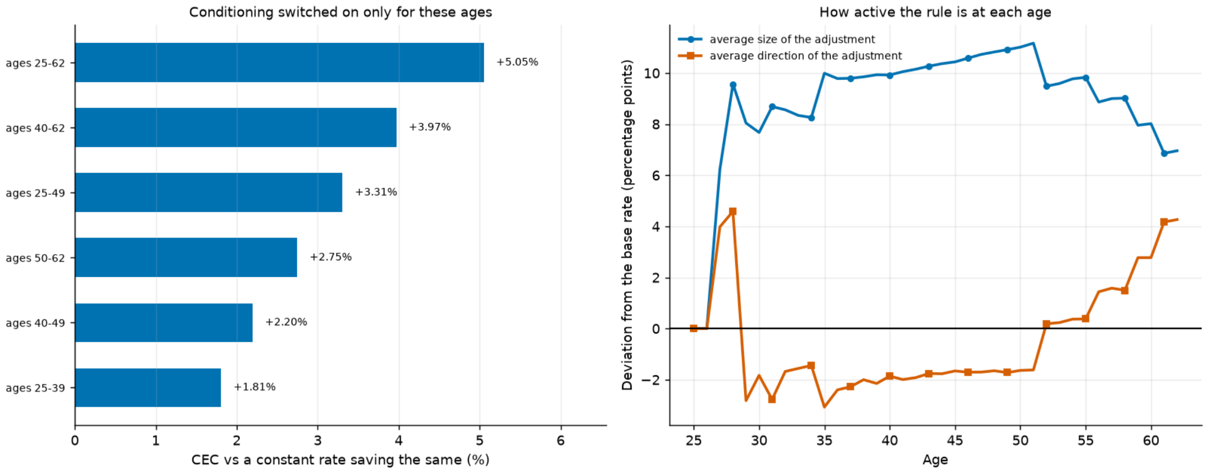

Figure 38.

Left: the value of conditioning when switched on only for the stated ages. Right: how far the rule moves the contribution at each age, and in which direction.

14.9 What the value depends on

Table 45. The value of conditioning by what the money is invested in

Portfolio	Net value (%)	Mean rate	P(ruin)
bills only	5.74	15.8%	55.9%
target date fund	4.66	10.4%	21.9%
sixty forty	4.60	11.8%	24.0%
balanced all equity	4.43	9.6%	11.7%

Notes. Each portfolio is scored against its own constant-rate frontier, so these are not contaminated by differences in the level of the outcome.

Conditioning is worth most to the investor whose plan is going worst. Across the four portfolios the value tracks the ruin probability almost exactly: bills-only, which ruins 56% of the time, gets +5.74%, while all-equity, which ruins 12% of the time, gets +4.43%. Note that this runs *opposite* to portfolio volatility: the safest portfolio benefits most. What the signal is worth is not driven by how much the balance bounces around, but by how often it ends up somewhere the investor cannot live on.

Table 46. The value of conditioning by how risky the pay cheque is

Income shock s.d. (× baseline)	Net value (%)	Mean rate	P(ruin)
0.5×	2.05	9.9%	11.7%
1×	4.43	9.6%	11.7%
1.5×	4.90	9.3%	11.7%

Notes. Labour-income shock standard deviations scaled up and down, with the target and the constant-rate frontier both recomputed for each arm.

Labour-income risk cuts the other way from the portfolio result, and in the direction the dispersion story predicts: more income risk means more dispersion in where a path ends up relative to target, and more for the signal to correct. Taken together, the two interactions say that the funded ratio is worth reading in proportion to how far off target a path can drift and how badly that drift ends — not in proportion to how much the portfolio moves year to year.

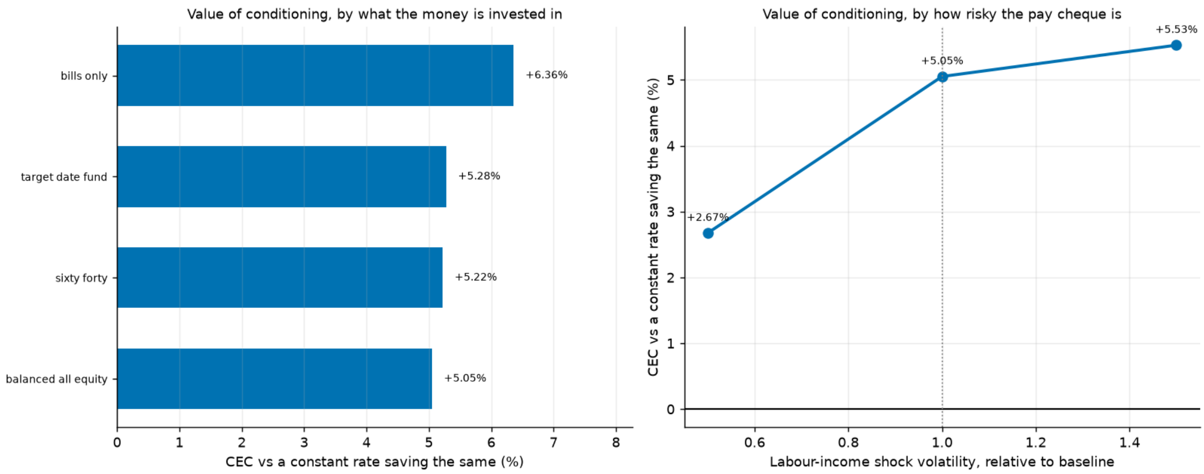

Figure 39.

The value of conditioning by portfolio and by labour-income volatility. The two point in opposite directions with respect to "risk", which is why the mechanism is about outcomes rather than variance.

15. Discussion

15.1 What the replication does and does not establish

The central claim reproduces cleanly and survives an unusually wide robustness exercise. That is a real result and it should update anyone whose prior rested on i.i.d. US-only simulation. But it is worth being precise about what has been established.

What has been established is a statement about a particular return-generating process: if the future resembles a block-resampled draw from the developed-market twentieth century, then a diversified all-equity portfolio dominates a declining glide path on essentially every metric a lifecycle investor would use. What has *not* been established is that the future will resemble that process. The bootstrap has no view on valuations, on the secular decline in real yields, or on whether the equity premium that the twentieth century delivered is a structural feature or a realised surprise.

That distinction matters most for the sceptic's strongest objection, which we take seriously: the sample is a sample of survivors at the level of the *system* even where it is not at the level of the country. Every country in the panel ended the century with a functioning capital market. A world in which that was not true is outside the support of the sampler, and no amount of block resampling can put it back in.

15.2 The hierarchy of levers

Reading the extensions together produces a ranking of what actually moves a retirement outcome, and it is not the ranking the industry's attention implies.

- **When you retire dominates.** The decade around the retirement date explains 6% of the variation in retirement consumption — more than the gap between any two allocation strategies tested here. It is also the one dimension no allocation rule can diversify.
- **How much you save dominates what you hold.** The savings-rate dimension of the tornado analysis moves the outcome by more than most allocation choices, and conditioning the savings rate is worth more than every allocation refinement in Sections 8 to 11 combined.
- **How you spend it matters more than expected.** The gap between the best and worst spending rules in Section 7 is comparable to the gap between the best and worst allocation strategies in Section 5.
- **What you hold matters, but mostly through diversification.** The all-equity advantage is real, and it is driven by the international leg rather than by equity exposure as such.
- **Currency hedging is close to irrelevant** at any realistic cost, the fine structure of the glide path is worth less than a basis point at most ages, and freeing every portfolio weight at every age adds less than the difference between two adjacent spending rules.
- **Leverage is barely a lever** at household borrowing costs. The advantage rounds to nothing across most of the plausible range of spreads, and what it does buy is bought out of the left tail. The one exception is worth noting: the leverage schedule is the only policy in this paper that an unconstrained search makes genuinely age-varying, and it declines steeply.

A reader looking for the highest-value change to their own plan should read that list from the top, not from the bottom. The profession's attention is allocated in roughly the reverse order.

15.3 Why the pay cheque beats the portfolio

The single most surprising result in the paper is that the best state variable for a savings rule is labour income relative to its expected path, not the portfolio balance. The mechanism is worth spelling out because it generalises.

Both signals identify the same underlying problem — this path is falling behind — but they differ in *when* the information arrives relative to the consumption decision. An income shock is observed at the moment income is received and before it has been committed to consumption; saving a larger fraction of a windfall costs almost nothing in utility because the counterfactual consumption was never planned. A portfolio shortfall is observed after the consumption plan is set, so responding to it means cutting consumption that the household expected to have. Under a concave utility function those two are not the same transaction.

The practical implication is unusually clean. "Save a fixed fraction of every raise and every bonus" is a rule that requires no target, no balance lookup and no arithmetic, and on this model it outperforms the wealth-target advice that dominates financial planning.

15.4 Implications for default design

If the results here were taken at face value by a plan sponsor, the implied redesign would not be "raise the equity share of the target-date fund", though that is what the headline suggests. It would be a reordering of what the default does at all.

- **Default the contribution schedule, not just the portfolio.** Auto-escalation tied to pay increases captures the strongest signal in Section 14 and requires no member decision.
- **Default the drawdown policy.** Section 7 shows the choice of withdrawal rule is worth as much as the choice of portfolio, and it is almost never defaulted.
- **Stop quoting four percent.** On this panel the sustainable rate at a five-percent ruin tolerance is 3.2% even on the best strategy tested. A default that plans around four percent is planning around a one-in-seven failure.
- **Treat the retirement date as a plan variable.** The largest single source of outcome variance is the one members are given the least help with.

We state these as implications of the model, not as advice. The model has no disutility of labour, no taxes, no fees, no housing and no behavioural constraints, and each of those would temper the conclusions in Section 16's direction.

16. Limitations and Threats to Validity

This section is deliberately long. A replication that reports only the ways in which it succeeded is not much use.

16.1 The cross-section is sixteen countries

This is the largest weakness in the paper, and it is a weakness we chose deliberately. The developed-market universe runs to 38 markets. We cover 16, because the other 22 have no recorded return series in any openly licensed source. Section 3.6 audits what we do use; Section 3.6.1 reports what the excluded countries have and why it is not enough.

Every statement in this paper about international diversification is therefore a statement about an average over 15 other developed markets, weighted by history length and drawn jointly with the domestic leg. That is a real limit on how far the mechanism generalises, and in particular the panel is entirely advanced-economy and entirely survivor: markets that closed and never reopened are not in the source database, so the tail this paper measures is the tail of markets that came back.

The last five years of the series are separately flagged as unverified in Section 3.6, on a variance test that every country in the sample fails in the same direction.

16.2 What the bootstrap cannot represent

- **No valuation conditioning.** Blocks are drawn without regard to the starting dividend yield or price-earnings ratio, so a simulated lifetime beginning at a market peak is statistically identical to one beginning at a trough. If long-horizon returns are predictable from valuations, the dispersion here is too wide and the mean too high.
- **No regime structure.** Block resampling preserves local persistence but not the possibility that the whole return-generating process changes. Secular declines in real yields, changes in the corporate tax base and the shift toward intangible capital are all invisible.

- **System-level survivorship.** Every country in the panel still had a functioning market in 2020. The set of histories in which that is false has zero probability under this sampler.
- **Blocks respect gaps, and gaps are informative.** A country with a wartime hole contributes fewer long blocks, so the sampler is mildly biased away from exactly the disrupted histories that motivate the exercise. We quantify this rather than correct it.

16.3 What the lifecycle model omits

- **No taxes.** Every return is pre-tax and every withdrawal is untaxed. Tax-deferred and taxable accounts behave differently under the same allocation, and the asymmetric treatment of capital gains and income would change the ranking of spending rules in Section 7 more than it would change the allocation ranking.
- **No fees.** A constant annual fee differential between an all-equity index and a target-date fund would move the results in the direction of the cheaper vehicle, which on current pricing is usually the index.
- **No housing.** For most households the primary residence is the largest asset and the mortgage the largest liability, and both interact with inflation in ways the financial portfolio does not. This is the single largest omission and it is not a small one. We do hold the data: Section 3.6.2 audits 1,805 country-years of observed housing returns and explains why an appraisal-smoothed index is not a fourth sleeve. That is a reason not to use it as written, not a reason the omission does not matter.
- **No economy-wide wage growth.** The income profile is an age effect only. Section 3.6.3 measures what that leaves out — a median 1.41% a year across 18 countries, which compounds to 1.68× over a career — and finds the bias runs against our conclusion, because less human capital weakens rather than strengthens the case for early equity.
- **No annuities.** A real annuity is the natural competitor to a withdrawal rule and is absent entirely.
- **No disutility of labour.** This is why the retirement-timing results of Section 12 require a matched comparison, and it means the model can never say when someone *should* retire, only what conditioning the date is worth.
- **Deterministic mortality.** Death at 93 with certainty. Real longevity risk is two-sided and would raise the value of the actuarial spending rules relative to the fixed-horizon ones.
- **No behavioural constraints.** Every rule here assumes the investor executes it. Section 14.6 prices the cost of a constrained contribution but nothing prices the probability that a saver abandons an all-equity portfolio in the middle of a 60% drawdown.

16.4 Specification sensitivities we know about

Three modelling choices are load-bearing and a reader should know where they bite.

- **The bequest shift.** Without De Nardi's κ, a single path with exactly zero terminal wealth sends the certainty equivalent to negative infinity at $\gamma \geq 1$. The value of κ is not identified by anything in the data and it changes the ranking of spending rules.
- **The evaluation window.** Allocation comparisons use the retirement window because working-life consumption is identical across strategies by construction; policy comparisons that change working-life consumption use the whole lifetime. Mixing the two would produce incoherent comparisons and we flag every switch.
- **The working-income floor.** Retirement carries a consumption floor through the progressive benefit schedule and working life, at zero, carries none. That asymmetry cancels in the allocation comparisons and does not cancel in the timing ones, where leaving it uncorrected inflates the value of retiring early by roughly a factor of two.

16.5 What the new searches assume

- **The simplex search is a coordinate search.** Section 9 reports restarts from three corners and they agree, but coordinate ascent offers no guarantee of a global optimum in a 204-dimensional space and none is claimed.
- **The four assets are the whole opportunity set.** There is no credit, no real estate, no commodities and no inflation-linked bond, and an inflation-linked bond in particular would change the fixed-income result of Section 9 more than any refinement within the sleeves that are present.
- **Leverage is modelled with limited liability** and a floating borrowing rate, rebalanced annually. Real margin lending liquidates at a threshold rather than at zero and reprices continuously, both of which would make borrowing less attractive than Section 10 finds it.
- **No borrowing constraint binds by quantity.** The model lets an investor borrow three times their financial capital at a spread; no lender would extend that against a retirement account.

16.6 Statistical caveats

Certainty equivalents are reported without standard errors. Under common random numbers the *differences* between policies are far more precisely estimated than their levels, which is what the comparisons rely on, but a reader wanting a confidence interval on any single level will not find one here. The optimisers are coordinate-ascent procedures on a grid: exact per coordinate under common random numbers, but with no guarantee of a global optimum in the joint space. Multiple restarts are reported where it matters.

17. Conclusion

We set out to reproduce a specific empirical claim and ended up with a hierarchy. The claim reproduces: on a 16-country panel of the developed-market twentieth century, sampled in blocks so that its persistence and its tails survive, a diversified all-equity portfolio delivers 11.2% more certainty-equivalent retirement consumption than a target-date glide path while failing less often. Solving for the allocation schedule directly rather than testing candidates returns the same answer, and the ranking holds across every parameter dimension we sweep. Freeing all four portfolio weights at every age — 204 parameters on the simplex, with nothing held fixed — reaches the same shape again and adds 0.30%; relaxing the long-only constraint pays well only when credit is nearly free, and breaks even by a 3.00% spread over the real bill rate.

But the extensions matter more than the replication. The single decade around a person's retirement date explains 6% of the variation in their retirement outcome — more than the entire allocation question. Conditioning the savings rate on financial position is worth more than every allocation refinement we test. And when we decompose that signal properly, the best state variable available to a saver turns out not to be their portfolio at all but their own pay cheque relative to what they expected to earn.

Several of the results here came out against the intuition that motivated them, and we have tried to report those cases with their diagnostics rather than smoothing them into a tidier story: the glide-path structure that dissolved under a deviation profile; the retirement-timing premium that halved under a matched baseline; the savings conditioning that turned out not to be tail insurance; the bull-market warning that the data did not support. A replication is most useful when it reports what it found rather than what it expected, and the machinery for telling those apart — matched baselines, common random numbers, deviation profiles, grid-edge checks — is as much a contribution here as any individual number.

The practical summary is short. On this evidence, an investor's attention is best spent, in order, on when they retire, on how much and when they save, on how they spend the portfolio down, and only then on what it holds — and when they get to what it holds, the answer is more diversified equity than the default gives them, for longer than the default holds it.

References

Anarkulova, A., Cederburg, S., and O'Doherty, M. S. (2023). "Beyond the Status Quo: A Critical Assessment of Lifecycle Investment Advice." Working paper.

Anarkulova, A., Cederburg, S., and O'Doherty, M. S. (2022). "Stocks for the Long Run? Evidence from a Broad Sample of Developed Markets." *Journal of Financial Economics*, 143(1), 409–433.

Ayres, I., and Nalebuff, B. (2010). *Lifecycle Investing: A New, Safe, and Audacious Way to Improve the Performance of Your Retirement Portfolio*. Basic Books.

Bengen, W. P. (1994). "Determining Withdrawal Rates Using Historical Data." *Journal of Financial Planning*, 7(4), 171–180.

Bodie, Z., Merton, R. C., and Samuelson, W. F. (1992). "Labor Supply Flexibility and Portfolio Choice in a Life Cycle Model." *Journal of Economic Dynamics and Control*, 16(3–4), 427–449.

Cooley, P. L., Hubbard, C. M., and Walz, D. T. (1998). "Retirement Savings: Choosing a Withdrawal Rate That Is Sustainable." *AAII Journal*, 20(2), 16–21.

De Nardi, M. (2004). "Wealth Inequality and Intergenerational Links." *Review of Economic Studies*, 71(3), 743–768.

Epstein, L. G., and Zin, S. E. (1989). "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework." *Econometrica*, 57(4), 937–969.

Guyton, J. T., and Klinger, W. J. (2006). "Decision Rules and Maximum Initial Withdrawal Rates." *Journal of Financial Planning*, 19(3), 48–58.

Jordà, Ò., Knoll, K., Kuvshinov, D., Schularick, M., and Taylor, A. M. (2019). "The Rate of Return on Everything, 1870–2015." *Quarterly Journal of Economics*, 134(3), 1225–1298.

Jordà, Ò., Schularick, M., and Taylor, A. M. (2017). "Macrofinancial History and the New Business Cycle Facts." *NBER Macroeconomics Annual*, 31, 213–263.

Merton, R. C. (1969). "Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case." *Review of Economics and Statistics*, 51(3), 247–257.

Politis, D. N., and Romano, J. P. (1994). "The Stationary Bootstrap." *Journal of the American Statistical Association*, 89(428), 1303–1313.

Samuelson, P. A. (1969). "Lifetime Portfolio Selection by Dynamic Stochastic Programming." *Review of Economics and Statistics*, 51(3), 239–246.

Waring, M. B., and Siegel, L. B. (2015). "The Only Spending Rule Article You Will Ever Need." *Financial Analysts Journal*, 71(1), 91–107.

Appendix A. Model Parameters

Every parameter below is read from a single YAML configuration file. No value is hard-coded in the simulation modules, which is what makes the sweeps of Sections 6, 11 and 12 mechanical rather than manual.

Table 47. The complete baseline parameter set

Parameter	Symbol	Value	Where it is used
Start age	—	25	All
Retirement age	—	63	All; varied in §6.4 and made endogenous in §12
Death age	—	93	All; varied in §6.4
Savings rate	s	10%	All; solved in §13 and conditioned in §14
Income profile (linear)	b_1	0.045	Labour income
Income profile (quadratic)	b_2	-0.0009	Labour income
Permanent shock s.d.	σ_p	0.10	Labour income; scaled in §14.9
Transitory shock s.d.	σ_t	0.25	Labour income; scaled in §14.9
Social-security formula	—	progressive	Retirement income floor
PIA bend point 1	—	0.21	× economy-wide average earnings
PIA bend point 2	—	1.28	× economy-wide average earnings
PIA replacement rates	—	90% / 32% / 15%	Successive tranches of career-average earnings
Withdrawal rule	—	fixed_real_rule	Baseline; eight families compared in §7
Withdrawal rate	—	4%	Baseline; swept in §5.4 and §6.4
Discount factor	β	0.96	Utility
Risk aversions	γ	2, 5, 10	Baseline $\gamma = 5$
Elasticities of substitution	ψ	0.5, 1.5	Epstein–Zin only
Bequest weight	θ	2	Utility; pivoted in §7.1
Bequest shift	κ	1	De Nardi (2004) specification
Consumption floor	—	0.0001	Numerical guard only
Evaluation window	—	retirement	Allocation comparisons; whole lifetime in §12–§14
Paths per strategy	N	100,000	All
Horizon	H	68	All
Mean block length	—	10 years	Bootstrap; swept in §4.1
Block length range	—	1–40 years	Bootstrap
Country draw	—	per_lifetime	Bootstrap; varied in Appendix C
Country weighting	—	history	Bootstrap; varied in Appendix C
Master seed	—	20240115	Reproducibility

Notes. Section references indicate where each parameter is varied. Parameters specific to a single extension (hedging cost, retirement trigger multiple, savings-rate grids, accumulation response grids) are documented in the configuration file and in the relevant section.

Appendix B. The Country Panel

The full panel, with real domestic equity statistics for each. Every country listed carries complete Jordà–Schularick–Taylor equity, bond and bill total returns over the years shown.

Table 48. Real domestic equity returns, every country in the panel

Country	Years	From	To	Mean	Geo. mean	S.d.	Skew	Kurtosis	AR(1)
Australia	118	1900	2020	7.8%	6.4%	16.8%	-0.31	0.4	-0.07
Belgium	125	1890	2020	5.9%	3.8%	20.7%	-0.00	0.0	0.08
Denmark	130	1890	2020	7.5%	6.1%	17.2%	0.43	1.8	0.04
Finland	125	1896	2020	8.9%	5.0%	30.0%	1.21	5.2	0.18
France	124	1890	2020	3.6%	1.1%	23.3%	0.86	3.7	0.15
Germany	124	1890	2020	8.7%	4.6%	28.9%	1.08	4.5	-0.01
Italy	124	1890	2020	6.1%	2.6%	26.8%	0.66	2.9	-0.04
Japan	129	1890	2020	7.9%	4.2%	25.8%	0.26	1.7	0.12
Netherlands	121	1900	2020	7.1%	5.1%	21.2%	0.74	2.9	0.03
Norway	131	1890	2020	5.7%	3.7%	20.1%	0.46	1.6	0.02
Portugal	131	1890	2020	3.2%	-0.1%	26.0%	0.66	2.4	0.25
Spain	115	1900	2020	6.2%	4.2%	21.1%	0.77	2.1	0.32
Sweden	131	1890	2020	8.1%	6.1%	20.3%	0.05	0.5	0.13
Switzerland	120	1900	2020	6.8%	5.1%	18.9%	0.30	0.4	0.10
United Kingdom	131	1890	2020	6.6%	4.9%	18.7%	0.90	5.7	-0.05
United States	131	1890	2020	8.4%	6.7%	18.7%	-0.24	0.1	0.00

Notes. Annual real total returns deflated by each country's own consumer price index. Kurtosis is excess kurtosis.

B.1 Structural gaps

The 16 contiguous runs of missing data in the panel's countries, preserved as gaps rather than interpolated. The bootstrap never draws a block that crosses one.

Table 49. Structural gaps in the empirical panel

Country	From	To	Years	Missing series
Australia	1890	1899	10	bond
Australia	1945	1947	3	bill
Belgium	1914	1919	6	bond, bill
Switzerland	1890	1899	10	dom_eq, bond
Switzerland	1915	1915	1	bond
Germany	1923	1923	1	bill
Germany	1944	1949	6	bond, bill
Denmark	1915	1915	1	bond
Spain	1890	1899	10	dom_eq, bond
Spain	1936	1940	5	bond, bill
Spain	2018	2018	1	bill
Finland	1890	1895	6	dom_eq
France	1915	1921	7	bill
Italy	1915	1921	7	bill
Japan	1946	1947	2	dom_eq
Netherlands	1890	1899	10	dom_eq

Notes. Concentrated around the two world wars, which is precisely the period a survivorship-prone sample would drop.

Appendix C. Supplementary Tables

C.1 Bootstrap sampling variations

The two structural choices in the sampler — drawing the country once per lifetime rather than per block, and weighting countries by history length rather than uniformly — are varied here. Neither reverses any ranking.

Table 50. Redrawing the country at every block

Strategy	Strategy	Cec Γ=5	P(Ruin)	Median Bequest
50/50 domestic/international equity	50/50 domestic/international equity	1.046	11.8%	43.028
100% domestic equity	100% domestic equity	0.878	25.9%	14.222
100% international equity	100% international equity	1.105	9.1%	63.743
60/40 domestic equity/bonds	60/40 domestic equity/bonds	0.866	24.9%	9.002
Target-date fund (glide path)	Target-date fund (glide path)	0.939	22.8%	8.726
100% bills (cash)	100% bills (cash)	0.737	57.5%	0.000

Table 51. Weighting countries uniformly rather than by history length

Strategy	Strategy	Cec Γ=5	P(Ruin)	Median Bequest
50/50 domestic/international equity	50/50 domestic/international equity	1.047	11.8%	44.715
100% domestic equity	100% domestic equity	0.881	25.4%	16.672
100% international equity	100% international equity	1.109	9.0%	64.113
60/40 domestic equity/bonds	60/40 domestic equity/bonds	0.870	24.2%	10.156
Target-date fund (glide path)	Target-date fund (glide path)	0.942	22.3%	9.336
100% bills (cash)	100% bills (cash)	0.739	56.4%	0.000

C.2 The income and benefit schedules

Table 52. The deterministic labour-income profile

Age	Real income	Multiple of age-25 income
25	1.000	1.000
30	1.224	1.224
35	1.433	1.433
40	1.604	1.604
45	1.716	1.716
50	1.755	1.755
55	1.716	1.716
60	1.604	1.604

Notes. Before permanent and transitory shocks. The hump peaks in the early fifties, which is what makes the solved savings profile of §13.2 hump-shaped at moderate risk aversion.

Table 53. The progressive social-security benefit schedule

Career-average earnings	Annual benefit	Replacement rate
0.307	0.276	90.0%
0.628	0.388	61.8%
0.949	0.491	51.7%
1.270	0.593	46.7%
1.591	0.696	43.8%
1.911	0.799	41.8%
2.232	0.856	38.3%
2.553	0.904	35.4%
2.874	0.952	33.1%
3.195	1.000	31.3%
3.516	1.048	29.8%
3.837	1.096	28.6%

Notes. US primary-insurance-amount bend points. The falling replacement rate is what supplies a genuine real consumption floor; a flat rate would scale with the outcome and insure nothing.

C.3 Long-horizon bootstrap outcomes

Table 54. Distribution of 68-year annualised and cumulative real returns

Series	Mean	1st pct	5th pct	Median	95th pct	99th pct	1st pct cumulative	Median cumulative
Domestic equity	4.43%	-5.02%	-1.53%	4.82%	8.89%	10.70%	-0.97	23.5
International equity	6.61%	1.66%	3.26%	6.65%	9.80%	11.21%	2.07	78.8
Bonds	1.33%	-6.68%	-3.29%	1.65%	4.66%	6.07%	-0.99	2.0
Bills	0.24%	-7.22%	-4.01%	0.69%	2.84%	3.67%	-0.99	0.6
Inflation	4.62%	0.80%	1.59%	3.96%	10.05%	13.51%	0.72	13.0

Notes. Annualised figures are geometric. The first-percentile cumulative multiple for domestic equity is below zero in real terms — a lifetime of holding a single national market that ended with less purchasing power than it started with. That outcome is the reason the international leg matters.

Appendix D. Computational Design

Every number, table and figure in this paper is produced by one program from the raw source files, and the document itself resolves its quoted figures from that program's output when it is typeset. Nothing is transcribed by hand. A rerun that changed a result would change the paper rather than leave it silently contradicting its own evidence.

This appendix describes how the computation is organised and how its correctness is established, because both bear on how much weight the results can carry.

D.1 Structure

The work is split into stages that each own one idea, so that an extension can reuse the baseline machinery instead of reimplementing it. Reuse is the point: every extension in this paper is compared against a baseline, and a comparison between two different implementations of the same model measures the implementations as much as the economics.

Table 55. How the computation is organised

Stage	What it owns	Paper section
Panel construction	Deflation to real returns, the leave-one-out international leg, currency-hedged legs, coverage and gap diagnostics, per-cell provenance	§3
Sampler	The stationary block bootstrap with gap-respecting admissibility; moment and persistence diagnostics	§4.1
Lifecycle simulator	The wealth recursion, income process, social-security schedule and strategy definitions	§4.2
Preferences	CRRA and Epstein–Zin certainty equivalents, the bequest term, shortfall and ruin metrics	§4.3
Sweep engine	Parameter sweeps under common random numbers; tornado and crossover analysis	§6
Spending rules	Eight families of withdrawal policy behind a single interface	§7
Glide-path solver	A batched evaluator and coordinate ascent over the age-by-asset schedule	§8
Allocation solver	The four-asset simplex solved at every age: lattice search then pairwise-exchange ascent	§9
Leverage	A levered evaluator, the cost-of-credit sweep and the age-varying leverage schedule	§10
Hedging	Covered-interest-parity hedged legs, break-even cost and optimal hedge ratio	§11
Path-dependent engine	Endogenous retirement dates and state-conditioned saving	§12–§14
Savings rules	Rule families, the fixed-mean shape solver and matched-rate scoring	§13–§14

D.2 Performance

Portfolio returns for a whole cohort are formed as a batched matrix product and the wealth recursion runs vectorised across paths. That is what makes a hundred thousand lifetimes per strategy — and the tens of thousands of policy evaluations the optimisers require — tractable on a single core, which in turn is what makes the sensitivity analysis of Section 6 affordable enough to run exhaustively rather than selectively.

D.3 Verification

Correctness rests on 523 automated tests. The ones that carry the most weight are equivalence tests between simulators, because every extension here is built on an engine that must agree with the one that produced the baseline:

- The path-dependent engine reproduces the fixed-date engine *bit for bit* given a fixed retirement age and a constant savings rate.
- The batched glide-path evaluator reproduces the reference simulator to a stated floating-point tolerance.

- Zero-sensitivity conditioning rules reproduce their own base age profile exactly, which is what makes the incremental values in §14 meaningful.
- The block sampler never draws across a data gap, verified against the run-length structure directly.
- A savings rule is never handed a return it has not yet lived through — the no-lookahead property is asserted, not assumed.
- Every available cell of the panel is an observation, checked against a per-cell record of where each number came from.

A further discipline runs through the prose. Where this paper states a direction — that one quantity exceeds another, that a ranking survives, that an effect has a sign — the statement is classified from the computed table when the document is typeset rather than written down in advance. The mechanism exists so that a result which moves takes its description with it, rather than leaving prose that quietly contradicts the table beneath it.

D.4 List of figures

D.5 List of tables